

Learning When to Change: Symbolic Supervision for Selective Evidence Sensitivity in Medical Reward Models

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Abstract

Reward models decide which medical reasoning traces are kept, reranked and reinforced. A useful reward reverses its preference when decisive patient evidence changes and holds it when similar-looking evidence does not apply: a penicillin allergy in the patient is a reason to change an order, the same allergy in the patient’s mother is not. We test this *selective sensitivity* with symbolic triplets from executable, stated clinical rules: a base case, a flip of one decisive input, and a near-miss that mentions the same concept without making it applicable. On rules of held-out structure, released medical PRMs solve 10–12% of triplets. Trained on decisive edits alone, a verifier reverses (92–93%) and does not hold (51–65%); near-misses in the same budget raise its triplet accuracy from 53.7% to 91.8%, and a separately written evidence record to 98.5–99.2%, a gain that a single applicability bit carries. The trained verifiers follow edits to the rule itself, but where no rule is stated they do not transfer: on MedE-inst and on exam-question pairs they are level with or below the untrained backbone. Selectivity is relative to a criterion. We therefore make the criterion a sourced input, taken from published score definitions, registered eligibility criteria, a diagnostic knowledge base and open ontologies, execute the parts of applicability that are executable, and report transfer over absent, stated, derived and wrong criteria (not yet run). A policy optimised against a released medical PRM that fails the test drives its reward to 1.00 while its accuracy falls from 88.0% to 38.7%.

1 Introduction

A reward model for medical reasoning is asked whether a step or an answer is right for the patient at hand. Often the answer turns on one line of the case. Under a rule that orders amoxicillin for streptococcal pharyngitis unless the patient is allergic to penicillin, the order is right for a patient with no

Stated rule. Strep. pharyngitis: order amoxicillin; if the patient is allergic to a penicillin, order azithromycin.

Claims. s : “Order amoxicillin.” s' : “Order azithromycin.”

Case	Reports	Rule	Required d
base x	allergies: none	s	> 0
decisive edit x^f	allergy: penicillin	s'	< 0
near-miss x^n	mother: penicillin all.	s	> 0
missing x^m	allergy status unknown	–	reject both
class member [†]	allergy: ampicillin	s'	< 0
class near-miss [†]	allergy: sulfamethoxazole	s	> 0

Figure 1: One rule, generated cases (schematic). $d = u(s) - u(s')$ is the preference a verifier should show. A verifier that prefers s throughout fails to reverse; one that prefers s' whenever the allergy is named reverses on the near-miss. [†]Class-level rows need a membership source (§3.2) and are not part of the frozen library.

drug allergy and wrong for one with a documented penicillin allergy (Figure 1). It is right again for a patient whose mother has that allergy. A reward that prefers the same order in all three cases fails to reverse when required; one that changes its preference in the second and the third reverses under an edit that leaves the decision unchanged. Whether evidence applies also depends on the rule: a stroke ten years ago counts under a score that asks for any prior stroke and not under a criterion that asks about the last six months.

Both failures matter because reward models filter training traces (Chen et al., 2024), rerank candidates (Snell et al., 2025; Yun et al., 2025) and supply the reward in reinforcement learning (Shao et al., 2024; Guo et al., 2025), where a learned reward that can be satisfied without solving the task is exploited (Gao et al., 2023; Tiwari et al., 2026). The first failure is documented: the mean accuracy of 28 LLMs on guideline-derived medication questions falls from 63.6% to 45.1% when a patient fact changes the answer (Hou et al., 2026). The second is easy to overlook: a change after a targeted edit means little unless it exceeds the change after

an edit that should not matter (Yang et al., 2026; Chen et al., 2026a). Medical reward models are evaluated on answer selection and injected step errors (Yun et al., 2025; Wu et al., 2026), neither of which changes the case. What is not established is what a reward learns when the “irrelevant” evidence concerns the very concept the rule tests, which supervision teaches it when that evidence applies, and what it needs where no rule is written down.

We build the test from executable rules (§3). Each rule yields *triplets*: a base case, a *decisive edit* (flip) of one input, and a *near-miss* in which the decisive concept appears without counting under the rule: in a relative, negated, at the wrong time, or as a value that approaches a threshold without crossing it. The rule program labels every case against the two claims the rule can support. Labels state what a stated rule implies, not what is clinically best. A *claim verifier* scores a claim against a case, a *process reward model* ranks traces with such scores, and a *training reward* is optimised against by a policy; a result at one level is not evidence for the next. Our contributions:

(1) **A test.** Triplets separate two failures that pair tests conflate, and rule-side edits change whether the same evidence counts while the case stays fixed. On rules of held-out structure, two released medical PRMs solve 10.0% and 11.9% of triplets and the untrained backbone 27.1% (§6.1).

(2) **A controlled comparison** of training distributions and representations under one budget (§6.2). Every corpus without near-misses yields reversal without holding. Near-misses raise the triplet accuracy of a verdict-only verifier from 53.7% to 91.8%, and a record written in a separate pass to 98.5–99.2%. A reader that writes one applicability bit matches the full record: what is learned is the applicability decision, not a schema.

(3) **A boundary, and a design that follows from it.** The trained verifiers follow edits to the applicability clause of the rule (85–94% crossed accuracy, against 39% untrained) and do not transfer where the criterion is implicit (§6.4): selectivity is relative to a criterion. We treat applicability as a conjunction of five typed conditions, each checked by an open symbolic source where one exists, and make the criterion a sourced input (*criterion grounding*, §4), to be evaluated on real clinical text with stated rules and released labels (§6.5; not yet run).

(4) **Reward use.** On exam questions, which state no criterion, the verifier’s score carries almost no

signal. As a training reward, a released medical PRM that fails the triplet test is exploited: its reward reaches 1.00 while accuracy on held-out rules falls from 88.0% to 38.7% (§6.6).

2 Related Work

Evaluator validity and contrastive supervision.

A valid judge is invariant to construct-preserving edits and sensitive to construct-changing ones (Chen et al., 2026a; Weng et al., 2026; Yang et al., 2026). Clinical tests change a patient fact that decides the answer (Hou et al., 2026; Chen et al., 2026b; Kim et al., 2025) or add clinically irrelevant distractors (Vishwanath et al., 2025); none pairs a decisive edit with an inapplicable mention of the same concept, edits the applicability clause of a rule, or is applied to medical PRMs. Label-flipping minimal edits (Kaushik et al., 2020; Gardner et al., 2020) teach which feature matters and not when it does not (Joshi and He, 2022); adding label-preserving edits removes over-triggering for negation scope (She et al., 2023) and safety policies (Liu et al., 2026b), and reward models are trained with causal and neutral augmentations (Srivastava et al., 2026; Liu et al., 2026a), whereas clinical post-training on guideline functions keeps only outcome-changing interventions (Shen et al., 2026). Our negatives keep the decisive concept and remove one condition of its applicability, a near miss in the sense of Winston (1970).

Symbolic verification, stated rules and sources of criteria.

PRMs are trained on step labels, roll-outs or generated verification (Lightman et al., 2024; Wang et al., 2024; Khalifa et al., 2025; Zhao et al., 2025), or on labels from formal tools (Kamoi et al., 2026). NS-PRM lets a deterministic verifier decide what is executable and trains the PRM on negatives that pass it (Zi et al., 2026); all its perturbations change the label, and we add the label-preserving side. Medical verifiers condition on retrieved passages (Yun et al., 2025; Zhang et al., 2026); scoring from a record written in a separate pass follows Dhuliawala et al. (2024) and Wang et al. (2026), with the assertion attributes of clinical NLP as fields (Chapman et al., 2001; Harkema et al., 2009). Rule-conditioned reasoning has many benchmarks (Saeidi et al., 2018; Holzenberger et al., 2020; Clark et al., 2020; Sun et al., 2024), yet deleting or replacing the rule left every compliance detector audited by Sadhu et al. (2026) unchanged. Clinical criteria are executed as

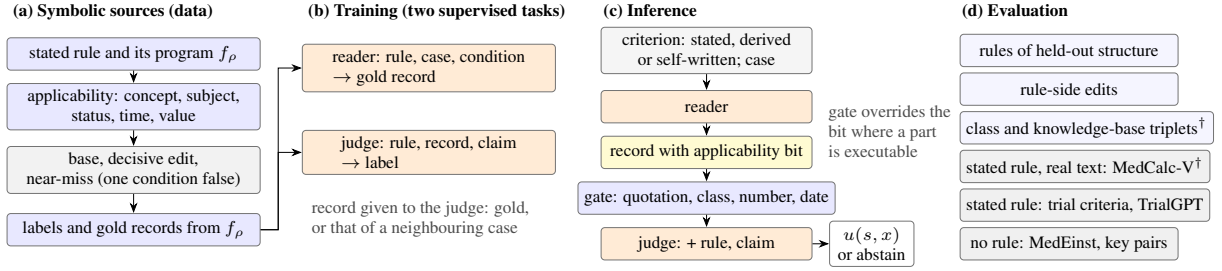


Figure 2: Overview. (a) A rule program, and for class terms an ontology, decide which mentions count; a near-miss falsifies exactly one applicability condition. (b) The program supervises two tasks. (c) At inference the criterion is an input with a source, and a symbolic gate recomputes the executable parts of applicability. (d) Controlled tests (blue) and external labels (grey). [†]Not yet run; the measured system is the reader and judge of (b) without gate.

guideline functions (Cao et al., 2026), checked as disease profiles (Cao and Chen, 2026) or compiled for a solver (Zhou et al., 2026), with prompted readers. Ontologies supply exactly labelled subsumption data with sibling negatives (He et al., 2023, 2024; Qu and Färber, 2026); injected into weights they changed medical QA little (Liu et al., 2025b), while a type vocabulary used as a gate before a learned judge helped (Shan et al., 2026). Learned rewards are exploited under optimisation (Skalse et al., 2022; Mahmoud et al., 2026; Pronesti et al., 2026) (Appendix I).

3 Rule Triplets, Criterion Sources and External Tiers

3.1 Rule tier

Rules and applicability. A rule ρ has criteria $c_1, \dots, c_m$ over typed inputs. A case is rendered from a state, a set of *mentions*, each with a concept, a value, a subject (patient or other), a status (present, absent, unknown) and a time (current or past). Every criterion c_j comes with an *applicability predicate* that says which mentions it counts, a conjunction over the five attributes $\mathcal{K} = \{\text{concept, subject, status, time, value}\}$:

$$A_j(m) = \bigwedge_{k \in \mathcal{K}} [m_k \in D_j^k], \quad (1)$$

where D_j^k is the set of values of attribute k that c_j accepts. The predicate is part of the rule: a creatinine threshold counts the patient’s current value, a stroke-risk score a stroke at any time. In 30.5% of criteria a past or family finding counts, so no global heuristic about time or subject is correct. In the frozen library the concept condition is an identity test. Rules are clinical scores or *constraint rules* (choose a unless c holds, then b). Each group is scored on *conclusion claims*, the two outcomes

the rule can support in neutral wording, and on *criterion claims* (“criterion j is met”). The prompt states the rule.

Edits. The engine edits one input of a criterion and keeps an edit only if executing the program confirms its invariant on the target claim. A *flip* changes a counted mention so that the claim changes. A *near-miss* reports the concept and value of the flip in a mention for which exactly one condition of Eq. 1 is false, and leaves the claim unchanged: another *subject*, an absent *status*, a *time* the predicate does not count, or a *value* that moves toward the threshold without crossing it (exactly at the threshold: *boundary*). Presentation edits re-render the same state, and missing twins replace a decisive input by an explicit unknown (Appendix C). *Rule-side items* keep the case and change one applicability clause of the rule (“ever” against “within the last six months”; the patient against the patient or a first-degree relative). An item crosses two rules with three cases and is solved only if all six judgments are right (crossed accuracy, XA); the 400 items were built after the library was frozen, and no system is trained on them.

Triplets and metrics. A triplet crosses a base case x , its flip x^f and a near-miss x^n with the claims s (correct for x and x^n) and s' (correct for x^f). For a reward with score $u(s, x)$ let $d(x) = u(s, x) - u(s', x)$. The triplet is *solved* if $d(x) > 0$, $d(x^f) < 0$ and $d(x^n) > 0$; ties are failures. *Reversal* (Rev) is the share of base–flip pairs with $d(x) > 0$ and $d(x^f) < 0$, *Hold* the share of near-misses with $d(x^n) > 0$, and *triplet accuracy* (TA) the share of solved triplets, the primary summary; TA_c is TA on criterion claims. Three fair preference signs solve a triplet with probability $1/8$, so TA is read with base

Tier	Criterion source	Label basis
<i>Controlled edits, labels by execution</i>		
Rule triplets, rule-side items	our 353 stated rules	rule program
Class triplets [†]	RxClass, HPO, Mondo	class closure
KB triplets [†]	DDXPlus finding lists	exclusion procedure
MedCalc-V edits [†]	published score text	released values, rule code
<i>Released labels, criterion stated</i>		
MedCalc-V criteria [†]	published score text	released values, rule code
Registered criteria [†]	registry text; Chia, LCT annotation	compiled annotation
TrialGPT	trial criterion text	physician labels
<i>Released labels, no criterion stated</i>		
MedEinst	none (DDXPlus lists [†])	benchmark labels
Key pairs	none (self-written [†])	answer keys
NLI4CT-P	trial report section	expert labels

Table 1: Tiers, with the source of each criterion and the basis of its labels (case text: Table 12). [†]Not yet built or scored. MedCalc-V: the rule-based subset of MedCalc-Bench Verified; KB: knowledge base.

accuracy. A deterministic scorer that sees only the claim, only the case, or only whether the decisive concept is named solves none (Appendix C).

3.2 Criterion sources and external tiers

Program-derived labels establish compliance with the stated rule, not that the behaviour matters outside the generator (Appendix B). Every external tier uses released labels or labels computed from released annotations, and every criterion has a source that we did not write (Table 1); we commission no clinical annotation.

Criterion stated by the dataset. *MedCalc-V* is the rule-based subset of MedCalc-Bench (Khan-dekar et al., 2024) in its maintained release, at a tag that postdates two independent audits (Ye et al., 2025; Krohn-Grimberghe, 2026): 19 clinical scores and 380 test notes, 230 of them human-written case reports and vignettes. Each instance ships the text of the score and the value of every criterion input, so criterion-level labels follow from released values by rule code. In the second audit, appending the calculator specification to the prompt raised one model’s answer accuracy from 51.9% to 81.5%, the effect our premise predicts for a verifier. *Registered criteria* are single-condition eligibility criteria quoted from the registry, whose operator, value, unit, time window and negation come from published annotated corpora (Kury et al., 2020; Dobbins et al., 2022). The *TrialGPT* annotations (Jin

et al., 2024) give physician judgments for 1,015 patient–criterion pairs; in only 33 does the label make the patient ineligible.

Criterion derived from an open source. The cases of *MedEinst* (Chen et al., 2026b) are narrated from DDXPlus patients (Fansi Tchango et al., 2022), whose released knowledge base lists which of 223 findings are associated with each of 49 conditions. For a pair of diagnoses a frozen script renders the findings listed for one only, the other only and both, with the one procedure the lists support (exclusion by a finding listed for the other diagnosis only), identically for both members of a pair. Class-level terms (“an ACE inhibitor”) take their membership from open ontologies: drug classes from RxNorm ingredients and RxClass (Nelson et al., 2011), phenotypes from HPO (Gargano et al., 2024), diseases from Mondo (Vasilevsky et al., 2022); none requires a SNOMED CT or UMLS licence. There the flip names a member and the near-miss a relative outside the closure of the class (Figure 1): the concept condition of Eq. 1, which the frozen library never varies.

No criterion. *Key pairs* are two exam questions that share answer options and differ in the keyed answer, from MedQA (Jin et al., 2021) and CareQA (Arias-Duart et al., 2025); what decides between the answers is knowledge that no input states. *NLI4CT-P* (Jullien et al., 2024) edits statements about trial reports (Appendix D). These tiers are the boundary of the method, not its target.

4 Ledger-RM

Reader. The reader R_ϕ receives the case x , the criterion text ρ , and the *condition under test* ω : the criterion, or the set of candidate answers, on which the claims disagree. It does not receive a candidate’s asserted outcome, so the claims of a triplet are judged from one ledger $e = R_\phi(x, \rho, \omega)$. An entry records facts and no decision: the condition the claim depends on (*need*), what the case says about it (*found*: a value, a verbatim quotation, or *missing*), and the subject, *status* and *time* of the finding. A quotation that does not occur in the case makes the ledger malformed. On the rule tier ω is supplied; an ablation has the reader find it.

Judge. The judge J_ϕ receives the ledger, the criterion text and the claim:

$$u(s, x) = J_\phi(R_\phi(x, \rho, \omega), \rho, s), \quad (2)$$

with $J_\phi(e, \rho, s) = \log p_\phi(+ | e, \rho, s) - \log p_\phi(- | e, \rho, s)$. The case reaches the verdict only through the ledger. A malformed ledger rejects both claims, and the tie counts as a failure. Reader and judge are two prompts of one adapter. Variants replace the ledger by prose that carries the same facts, drop the applicability fields, add or isolate a one-bit `applies` field written by the reader, or let the judge also see the case.

Training. On rules the program supplies the target ledger $e^*(x, \rho, \omega)$ and, for any ledger e , the verdict $y_\rho(e, s)$ obtained by executing the rule on the state that e describes. The reader is trained to write e^* . The judge is trained on a ledger that is e^* with probability $1 - p$ and otherwise the target ledger of another case of the same rule and condition, always with the verdict that the ledger shown implies, so no instance rewards a verdict that contradicts the evidence given to the judge (objective in Appendix H).

Symbolic gate. Three measurements (§6.3) motivate a composite system, Ledger-RM-G: the applicability decision is what the pipeline learns, the learned judge follows whole ledgers and not edits to single fields, and a rule program applied to the predicted ledger matches the judge. Ledger-RM-G therefore computes whatever part of Eq. 1 is computable. Each condition k has a check $g_k(e, c) \in \{1, 0, \perp\}$, with $\perp$ for not formalised or not normalised: a string match for the quotation, a comparison after unit conversion for a threshold, date arithmetic against a stated reference date for a window, membership in the closure of a class once the mention is linked to an ontology term by retrieval, and set membership for subject and status. If no check returns $\perp$, the conjunction replaces the reader’s `applies` bit; otherwise the bit stands and the judge also receives the case. With no criterion, or nothing to record, the system abstains and falls back to a verdict-only score. A criterion never has to be compiled in full: the gate acts on the parts that carry a number, a date or a class (Appendix F).

Criterion grounding. Where the task states no rule, a *grounding* function maps the task identity τ (the name of a score, a registered criterion, a pair of diagnoses, an option set) to a criterion text with a provenance tag, $g(\tau) = (\rho, \pi)$, $\pi \in \{\text{stated, derived, self-written}\}$. It never sees the case or its label, the members of a pair receive identical text, and its source is frozen before any

system is scored (Appendix E). Results are reported as a ladder over criterion conditions: *none*, *stated* or *derived*, *self-written* by the frozen backbone from τ alone, and *wrong* (another item’s criterion, or the pair’s own with the two roles exchanged), which tests that the verifier reads the rule.

For candidate selection, errors internal to a step are scored by a separate step check, the released Med-PRM, and combined with the ledger score by the minimum of the calibrated scores (Appendix H).

5 Experimental Setup

Rules and shifts. The frozen library (`rule_v1`) has 353 rules: 43 scoring rules implemented from published definitions, 60 constraint rules compiled from public recommendations by the tree-then-program procedure of Shen et al. (2026), and 250 rules sampled from a typed grammar (Appendix B). Each rule has a structural signature (rule kind, operators, dependency depth, input types, applicability predicate), in 21 classes. The main test, **L2**, has 2,000 triplets from 103 rules of held-out signature classes; **L3-inv** uses rules with invented names. Rendering templates and cue phrases for relatives, negation and time are disjoint between training and every test.

Training. All trained models use Qwen3.5-9B with LoRA (Hu et al., 2022) of rank 64 for one epoch on 60,000 records. Four corpora differ only in the cases they contain. *Natural*: one case per group, the decisive condition rare. *Balanced*: one case per group, outcomes equally frequent. *Blocks*: the base case and the decisive edit of every group. *Triplets*: blocks in which a near-miss replaces the base case in half of the groups. Representations: two one-stage formats (verdict only; the ledger as a rationale followed by the verdict) and three two-stage formats with a case-blind judge (a prose summary rendered from the same facts as the ledger, a value ledger with `need` and `found` only, and the applicability ledger).

Systems and protocol. We audit the released Med-PRM (Yun et al., 2025), MedS³ PRM (Jiang et al., 2025) and FoVer PRM (Kamoi et al., 2026), and on subsets ThinkPRM-14B and GenPRM-7B. Training-free baselines on Qwen3.5-9B are the zero-shot critic, default correction (Shi et al., 2024), a prompted summary, a prompted ledger, and a verifier that writes and executes a program for the rule

	Rule tier (L2)					ME Key	
	Rev	Hold	TA	TA _c	XA	pair	pair
Reward signal							
<i>Released PRMs</i>							
MedS ³ PRM	22.6	16.4	10.0	41.5	14.0	25.1	39.5
Med-PRM	19.4	86.0	11.9	64.2	13.5	16.2	57.7
FoVer PRM	49.9	61.9	31.8	68.4	44.5	21.0	67.2
ThinkPRM-14B ^a	96.5	90.0	87.5	tbd	tbd	tbd	tbd
GenPRM-7B ^b	tbd	tbd	tbd	tbd	tbd	22.0	tbd
<i>Qwen3.5-9B, no training</i>							
Critic	38.5	32.2	27.1	83.9	39.0	24.2	79.8
+ default corr.	tbd	tbd	26.4	tbd	58.5	24.7	tbd
+ prompted summary	74.7	89.8	67.5	88.8	80.5	tbd	79.6
+ prompted ledger ^c	68.0	87.8	59.5	82.1	52.5	tbd	tbd
Generated program ^d	93.3	94.7	89.8	tbd	82.8	–	–

Table 2: Audit (%), each signal with its native readout. Rev, Hold, TA: conclusion claims on L2; TA_c: criterion claims; XA: rule-side items; ME, Key: both members right on MedEinst and MedQA key pairs. ^a200 triplets. ^b200 pairs. ^cLenient parsing (strict: 98.8% of ledgers malformed, TA 0.05). ^d1,000 triplets; executes code it writes. Larger open and closed judges: **not yet scored**.

(Ocampo Osorio et al., 2026); larger judges are **not yet scored** (Appendix H). Scores are verdict log-odds. Key cells have five seeds; other values are from seed 0. Paired differences use a cluster bootstrap over rules (patients for TrialGPT, label pairs for MedEinst). The analysis plan was fixed before the test sets were scored, **and a second stage was registered after its outcome and before any of its data were built** (Appendix A).

6 Results

6.1 Audit of existing reward signals

No released medical reward model comes close to solving the triplets, and they fail differently (Table 2). Med-PRM is right on 91.2% of base cases and 86.0% of near-misses and reverses on 19.4% of flips: it holds because it does not respond. The MedS³ PRM prefers the wrong claim on 74.6% of base cases; three fair coins would solve 12.5%, which is why TA is read with base accuracy. The untrained backbone separates the two kinds of claim: asked whether a criterion is met it solves 83.9% of triplets, and asked which outcome the rule supports 27.1%, with base accuracy below one half: its errors lie in applying unseen rule structures more than in reading the evidence. A prompted summary before the verdict raises its accuracy to 67.5%, a verifier that writes and runs a program for the rule reaches 89.8%, and ThinkPRM-14B, a general verifier that reasons before it scores, 87.5% on a subset of 200 triplets: the

Representation	Nat.	Bal.	Blocks	Triplets
<i>One-stage (verdict produced with the case in context)</i>				
Verdict only	58.0	49.1	58.2	91.3
five seeds			53.7 ± 5.4	91.8 ± 0.4
Rationale, then verdict	55.1	64.7	61.5	91.0
<i>Two-stage (case-blind judge)</i>				
Evidence summary (prose)	45.7	72.5	67.3	98.0
four and five seeds			64.9 ± 6.2	98.5 ± 0.7
Value ledger	43.0	71.9	70.7	85.7
Applicability ledger	77.2	49.4	75.3	99.2
five seeds			71.0 ± 10.7	99.2 ± 0.3

Table 3: Triplet accuracy on L2 (%) by training corpus and representation; Qwen3.5-9B, rule data only, 60k records. First line of each row: seed 0; seed rows: mean and s.d. A prose judge that also sees the case: 98.8 on triplets. Intervals over rules at seed 0 are at most ±4.9 points (Appendix H).

test is solvable without training on our rules, and the released medical PRMs do not solve it. Among shortcut scorers (Appendix G), a claim-only scorer and one that sees whether the decisive concept is named solve no triplet, and a trigger lexicon solves 54.0% with training cue phrases and 99.5% with the test cue phrases: the difficulty of the generated test lies in unseen wording and rule structure, which is why the rule-side items and the external tiers matter.

6.2 Training distribution and representation

Supervision without near-misses yields reversal without holding. In the fifteen cells of Table 3 trained without near-misses, reversal at seed 0 is between 83.2% and 99.6% and near-miss correctness between 43.9% and 77.8%. Over five seeds the verdict-only model trained on blocks reverses on 91.8–93.4% of flips, is right on 51.3–64.9% of near-misses, and solves 53.7% of triplets. On criterion claims it ends below where the backbone started (53.5–66.5% against 83.9%): a verifier that has only seen evidence that counts learns that a mention is a reason to change. Replacing the base case by a near-miss in half of the groups raises triplet accuracy to 91.8% (+38.2 points [34.9, 41.2]), with near-miss correctness of 96.6–98.2% at unchanged reversal, and the standard deviation over seeds falls from 5.4 to 0.4 points. The prose record moves from 64.9% to 98.5% and the ledger from 71.0% to 99.2%. Without near-misses the failures are concentrated where the finding belongs to another person (11.5% solved) or the value sits on the threshold (33.8%). Using near-misses only as probes for re-weighting (Liu et al., 2026a) gives 55.3%, and outcome balance is not the remedy: on balanced

corpora near-miss correctness stays at 51.6–72.8%.

What the second stage adds. With near-misses, every format except the value ledger solves at least 97.7% of triplets on criterion claims. They differ on conclusion claims under held-out rule structure: the one-stage formats reach 91.3% and 91.0%, the two-stage formats 98.0% and 99.2%. The rationale writes the same ledger in the same pass as the verdict and gains nothing over the verdict alone. What helps is a separately supervised reading pass and a second pass that applies the rule to a short record, not writing evidence first and not hiding the case. Typed fields add 0.8 points over prose with the same facts ([0.3, 1.3]); a ledger without subject, status and time is 13.5 points lower.

6.3 What the judge uses

One applicability bit carries the result. If the reader also writes whether the condition applies, a judge that sees only that bit and the claim solves 99.3% of L2 triplets, and a reader trained to write the bit and nothing else 99.95% (Table 21). Both reach 97.8% crossed accuracy on rule-side items, against 85.0–94.0% for the ledger: what is learned is the reader’s decision that a finding counts under the rule in view. On rule-side items that differ in a time window the bit reader solves 91% and the ledger 68%, because a field that says “current” or “past” cannot carry a date.

The learned judge follows whole ledgers, not fields. Given the ledger of another case of the same rule and condition, the judge returns the verdict that ledger implies in 99.9% of instances. When one of subject, status or time is edited in a correct ledger, it follows 47.8% of the edits that should change the verdict and keeps 99.7% of the verdicts that should stay: it is not a faithful function of the fields. A rule program applied to the predicted ledger gives 99.4%, against 99.2% for the learned judge (+0.2 [0.0, 0.6]). Where a rule is executable, what the pipeline contributes is the reading; hence the gate of §4.

Conditions not seen in training. Trained on triplets without one kind of near-miss, the verdict-only model solves 7.0% of L2 triplets of the kind “another person” (91.0% when trained with it), 60.2% for “past” (88.2%) and 49.8% for “threshold” (93.2%); the ledger 62.5%, 86.2% and 87.0% (Table 18). Near-miss supervision has to cover each condition of Eq. 1, which is the reason to hand a condition to a symbolic check where one exists (further ablations: Appendix H).

6.4 Transfer within and beyond rules

Within rules, and when the rule changes.

Trained on rule triplets only, Ledger-RM solves 99.2% of triplets on rules of held-out structure and 97.9% on invented rules; the verdict-only model 91.3% and 96.9% (Table 4, Table 22). On rule-side items, where the case is fixed and one clause of the rule changes, the triplet-trained verifiers solve 85.0–94.0% (Ledger-RM) and 85.5–91.2% (verdict only) over five seeds, the untrained backbone 39.0% and Med-PRM 13.5%. Unlike the compliance detectors of Sadhu et al. (2026), these verifiers read the rule. Time windows are the weakest clause (68% for Ledger-RM at seed 0, against 93–100% for the others): the frozen library contains none.

Stated criterion, expert labels. On the Trial-GPT annotations the systems trained on triplets have a higher macro-F1 than the backbone (69.1–70.8 against 65.7), and no difference is established: +3.4 points [−2.6, 9.2] for Ledger-RM, and −1.8 [−4.8, 1.1] against the same format trained on blocks. The gain comes from answering “not enough information” more often; on items with a decided label the two-stage systems are below the backbone (84.4–87.7% against 92.8%).

No criterion stated. Where the criterion is implicit, training on rules changes nothing or does harm. On MedEinst the untrained backbone is right on both members of 24.2% of pairs; with control accuracy of 71.8% and trap accuracy of 51.6%, it gives the same diagnosis to both members of 75.0% of pairs and is wrong on both in 0.8%. When it responds to the edited sentence it responds correctly, and it seldom responds. The verdict-only model trained on triplets behaves the same (76.8%; pair accuracy 22.8%), and Ledger-RM falls to 15.7%, with 35.9% of its reader outputs malformed on these long narratives. On key pairs the verdict-only models are level with the backbone (78.2–78.7% against 79.8%) and the two-stage systems collapse to 30.5–35.9%: a reader with no criterion has nothing to read against, and a judge with no case cannot supply the knowledge.

By the decision rule fixed in advance (Appendix A), the zero-shot medical claim of the first plan is withdrawn. The pattern is nonetheless specific: the trained verifiers are selective where a criterion says which evidence counts, and unchanged where none does.

System	Rule tier				Criterion stated		No criterion stated					
	L2	L3-inv	XA	Hold	TrialGPT		MedEinst			Key pairs		NLI4CT-P
					F1	choice	pair	control	trap	MedQA	CareQA	
<i>Training-free, Qwen3.5-9B</i>												
Critic	27.1	33.4	39.0	32.2	65.7	92.8	24.2	71.8	51.6	79.8	84.4	80.9
+ prompted summary	67.5	tbd	80.5	89.8	68.0	91.3	tbd	tbd	tbd	79.6	80.9	tbd
<i>Trained on rule data only (all clinical columns are zero-shot)</i>												
Verdict only, rule blocks	58.2	62.9	60.8	61.5	64.8	92.8	20.5	68.3	51.8	78.7	82.7	85.6
Verdict only, rule triplets	91.3	96.9	90.2	97.7	70.8	92.8	22.8	69.5	52.9	78.2	83.1	85.4
Evidence summary, rule triplets	98.0	97.6 ^a	88.5	99.3	69.2	87.5	25.1	73.2	47.3	33.3	55.6	73.2
Ledger-RM, rule blocks	75.3	70.1	76.0	76.2	70.9	85.8	tbd	tbd	tbd	35.9	44.0	tbd
Ledger-RM , rule triplets	99.2	97.9	89.8	99.95	69.1	87.7	15.7	44.4	35.1	30.5	38.2	tbd
<i>Reference that uses the rule program at test time</i>												
Program on the trained reader's ledger	99.4	–	–	100.0	–	–	–	–	–	–	–	–

Table 4: Transfer (%), seed 0, ledger score alone. L2, L3-inv: triplet accuracy on rules of held-out structure and on invented rules; XA: crossed accuracy on rule-side items; Hold: on L2. TrialGPT: macro-F1 over met, not met and not enough information against physician labels, and accuracy of the preferred claim on items with a decided label. MedEinst: both members right, and accuracy on control and on trap cases. Key pairs: both members right. NLI4CT-P: mean of control F1, faithfulness and consistency. ^aSeed 1. All rows: Table 20.

6.5 Criterion grounding

These experiments are specified and not yet run.

If selectivity is relative to a criterion, three things should hold, and each can fail. (i) On real clinical text with a stated rule and released labels, the ordering of the rule tier on criterion claims should reappear: triplets above the backbone, blocks below it. On MedCalc-V the verifier judges such claims on untouched notes; triplets change one number across or toward a threshold, or insert a sentence that gives the finding to a relative, negates it or places it in the past. (ii) Removing or exchanging the criterion should lower the score of the trained verifier by more than that of the backbone (Table 14). (iii) Where no criterion is stated, a derived one should help only as far as it decides the released labels. Before any model is scored on MedEinst with the derived criterion, a symbolic executor applies the stated procedure to every pair; its accuracy is what the criterion entails, and we expect it to be well below 100%, since trap labels were accepted as plausible, not as entailed. Triplets generated by script from DDXPlus test patients test the same source without that gap, and class triplets compare the learned verdict with the gate (Appendix E).

6.6 Reward use: selection and policy training

Selection. Reranking 16 samples of a frozen policy on exam questions tests the verifier as a process reward where no criterion is stated (Table 25). The expectation of the first plan, that adding the ledger score to a step check improves on the step check, is not supported (-0.9 points on MedQA $[-2.1, 0.2]$), and self-consistency is the best selector. As a trace

score for answer correctness the ledger score has an AUC of 0.51 (Med-PRM: 0.65); with no criterion in the input it has nothing to verify.

Policy training. A reward that fails the triplet test can be optimised without solving the task. We train a Qwen3.5-4B policy with GRPO for 1,000 steps on rule tasks and evaluate it with the rule program on 150 held-out L2 pairs (one seed; descriptive). With an outcome reward, accuracy on both members of a pair rises from 88.7% to 96.7%, and with a reference-graph reward after Mu et al. (2025) to 98.0%. With the released Med-PRM as the reward it falls from 88.0% to 38.7% while the mean reward reaches 0.9997; within a rollout group this reward is uncorrelated with correctness (-0.05), as its reversal rate of 19.4% would suggest. Runs with the two ledger rewards are **not finished**. Such exploitation is known (Tiwarei et al., 2026; Mahmoud et al., 2026); we add an exactly labelled held-out evaluation and the exploited reward's triplet profile.

7 Conclusion

On executable rules, medical reward models fail in two directions, and supervision on decisive evidence alone cures the first failure and produces the second. Near-misses in the same budget remove it; one learned bit, whether the finding counts under the rule in view, accounts for the gain. Such a verifier follows edits to the rule and is unchanged on benchmarks that state none: selectivity is relative to a criterion, which has to be an input with a source. Whether this carries onto clinical text is the experiment of §6.5 (not yet run).

649 Limitations

650 **What the labels are.** Rule-tier labels establish
 651 compliance with a stated rule, its applicability pred-
 652 icates and a rendering grammar. They say what the
 653 rule implies, not what is clinically optimal, and
 654 that a program produced a label does not show
 655 that the text expresses the state, that the test is
 656 free of generator artefacts, or that the behaviour
 657 matters outside the generator (Appendix B). We
 658 commissioned no clinical annotation, no clinician
 659 reviewed the rules, and the reading of rendered
 660 groups before the freeze was done by model ses-
 661 sions, not by people; reused benchmarks carry the
 662 review their authors report. The TrialGPT annota-
 663 tions are physician judgments on synthetic patient
 664 summaries, made with a model’s explanation in
 665 view; MedEinst cases derive from a simulator and
 666 were edited and filtered by models, with a physi-
 667 cian review of a sample reported by its authors;
 668 key pairs inherit the errors of exam keys and are
 669 not counterfactuals. Nothing here supports clinical
 670 use.

671 **Transfer.** The verifiers trained on rules do not
 672 improve agreement with any clinical benchmark
 673 that leaves its criterion implicit, and the improve-
 674 ment where a criterion is stated is not statistically
 675 established. The experiments that would show se-
 676 lectivity on real clinical text are specified and not
 677 run. If they fail, what remains is a controlled study
 678 of supervision for rule-conditioned verification.

679 **Sources of criteria.** A criterion derived from a
 680 knowledge base or an ontology is exact only rela-
 681 tive to that source and its release. Open knowledge
 682 bases mostly list associations, not decisions; class
 683 membership differs between drug classifications,
 684 so we keep a membership only where two sources
 685 agree, and a missing contraindication assertion is
 686 not evidence of safety. Giving a model the lists
 687 from which a benchmark was generated changes
 688 what the benchmark measures, which is why the
 689 ladder always includes the condition without a cri-
 690 terion and the untrained backbone under the same
 691 text. Self-written criteria can be wrong, and a rule-
 692 follower follows wrong rules.

693 **The rule language.** The test is generated from
 694 a finite language of applicability conditions. Held-
 695 out signature classes reuse primitives and operators
 696 seen in training, 76.3% of criterion subexpressions
 697 of the main test also occur in training rules, there
 698 are few such classes, and most judgments concern
 699 one condition of one rule. A trigger lexicon that

is given the test cue phrases solves the generated
 test, and near-miss supervision does not extend to
 a condition it has not covered (§6.3). The frozen
 library has no time windows and no class-level
 concepts.

The reader. The record assumes that a claim
 depends on a few identifiable findings. Claims that
 weigh many soft findings fit it poorly: on long
 diagnostic narratives over a third of the reader’s
 outputs are malformed. A quotation can exist in
 the case and still be the wrong evidence, the judge
 is not a faithful function of the fields, and on the
 rule tier the condition under test is supplied.

Verifier, process reward, training reward. The
 triplet results concern a claim verifier. As a selector
 on exam questions the ledger score is uninforma-
 tive. The policy experiment has one seed and one
 policy, and its runs with our own rewards are un-
 finished; we make no claim that a policy trained
 against our verifier is safer.

Scope. One backbone family; five seeds for the
 key cells and one elsewhere; no larger or closed
 judge has been scored; English only.

Ethical Considerations

The work uses public benchmarks and generated
 cases; no patient data were collected. Stated rules
 in our data are test specifications and must not be
 read as prescribing guidance, and criteria derived
 from knowledge bases or written by a model are
 not clinical recommendations. A reward model
 that appears to check patient evidence could be
 trusted more than it should be; accuracy on these
 triplets does not establish safety in care, and the
 models are released for research on post-training
 only. Because the engine generates cases on which
 a given reward fails, it could be used to construct
 inputs that mislead a deployed evaluator; we see
 this as an argument for auditing rewards before
 they are used. Redistributed items derived from
 MedCalc-Bench carry its share-alike licence, and
 ontology content is used under the terms of each
 source (Appendix E).

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A Analysis Plan and Its Outcome

Draft status. Values printed in black are measured on frozen sets (results of 3 October 2026); seed-0 values are marked as such. Everything printed in red is missing: “tbd” marks a value from an experiment that has not been run or whose result we have not received, and red text names a fact to be confirmed against the repository. This draft contains no target or example values. The red banner and this paragraph disappear, and the build fails on any remaining placeholder, once measured values are entered.

Stage 1 (fixed before any test set was scored). *Primary endpoints:* triplet accuracy on L2 and pair accuracy on MedEinst test pairs for the model trained on rule triplets only. *Primary comparisons,* paired on identical items with a cluster bootstrap: (1) balanced against natural, verdict only, L2; (2) triplets against blocks, verdict only, L2; (3) applicability ledger against prose summary, triplets, L2; (4) rule-only Ledger-RM against the untrained critic, MedEinst; (5) rule-only Ledger-RM against the rule-only prose pipeline, MedEinst; (6a) rule-only Ledger-RM against the untrained critic and (6b) against the same format trained on blocks, TrialGPT test portion (added before any external result was seen). *Decision rules:* if (2) fails, near-misses are reported as an evaluation device only;

if (3) fails, the typed ledger is reported as equal to prose; if (4) and (6) fail, the answer outside rules is negative and the medical claim is withdrawn.

	Difference (points)	<i>p</i>	Outcome
(1)	−8.8 [−11.9, −5.8]	<0.001	expectation not supported
(2)	+38.2 [34.9, 41.2]	<0.001	holds (five seeds)
(3)	+0.8 [0.3, 1.3]	0.004	holds, at 0.8 points
(4)	−8.5 [−20.1, 1.7]	0.104	fails
(5)	−9.4 [−19.5, −0.8]	0.026	fails (negative)
(6a)	+3.4 [−2.6, 9.2]	0.244	not established
(6b)	−1.8 [−4.8, 1.1]	0.214	not established

Table 5: Outcome of the seven primary comparisons of stage 1. Intervals for (4) and (5) resample the 134 label pairs; with pairs as units they are [−9.8, −7.0] and [−10.7, −8.1]. *p*-values are unadjusted; with the Holm correction over the family (to be recomputed by the table script) only (1), (2) and (3) remain below 0.05.

Application of the decision rules. (2) holds by a wide margin: near-miss supervision is reported as a training result. (3) holds at a size that does not carry a paper: the typed ledger is reported as 0.8 points above matched prose. (4) and (6) fail: the zero-shot medical claim is withdrawn, and the abstract and §6.4 say so. The expectation that combined selection improves on a step check, which was not a primary comparison, is not supported either (§6.6). **Stage 2 (written on 8 October 2026, after the outcome of stage 1 and before any stage-2 set was built; to be committed to the repository, with its hash, before the first stage-2 set is built).** The question is whether the boundary found in stage 1 is the absence of a criterion. *Primary comparisons,* Holm-corrected as one family: (7) MedCalc-V criteria on human-written notes, criterion stated: verdict-only verifier trained on triplets against the one trained on blocks (balanced accuracy over items that score points and items that score none, notes as clusters); (8) on the same items, the interaction of criterion (stated against none) with system (triplet-trained verifier against untrained critic); (9) MedCalc-V edits: triplet accuracy, triplets against blocks; (10) class triplets on held-out classes: gate against learned verdict; (11) MedEinst, all pairs: derived criterion against none, for the triplet-trained verdict-only verifier, against the same difference for the critic; (12) key pairs: case-visible composite against the verdict-only verifier, non-inferiority at a margin of two points. Comparisons (7), (8), (9) and (11) use the adapters of stage 1 without further training. *Decision rules:* if (7) and (9) fail, near-miss supervision

is a result about generated text and the abstract and conclusion say so. If (8) shows no interaction, the trained verifier’s advantage does not come from reading the criterion. If the symbolic executor decides fewer than half of the MedEinst pairs, (11) is dropped and MedEinst is reported only as the contrast without a criterion. If (10) fails, class conditions are left to the learned judge. If (12) fails, the composite is reported as harmful where no criterion is stated. *Selection and freezing:* prompts, thresholds and the linking pipeline are chosen on development portions (training notes of MedCalc-V, DDXPlus validation patients, MedEinst reference pairs, classes disjoint from the test classes); each stage-2 set is frozen with a hash before any system is scored on it, and `rule_v1` is not modified.

B Rule Library and Splits

Source	Rules
Published scores, our implementation	43
Constraint rules from public recommendations	60
Grammar-sampled rules	250
Total (<code>rule_v1</code> ; 21 signature classes)	353
Invented rules, test only (L3-inv)	60
MedCalc-Bench calculators	not included

Table 6: Rule population of the frozen library. Of the 103 rules written from sources, 97 are simplified source rules and 6 are synthetic; 53 deviate from their source in a recorded way.

What the labels establish. A program-derived label answers one of four questions (Table 7); the paper treats them separately.

Question	Evidence
Does the program implement the stated rule?	boundary, unit and dependency tests; target-claim check re-executed on every group (0 violations in 203,014 records)
Does the text express the intended state?	model reading sessions before the freeze (below); reading by the authors pending
Does the test measure applicability and not artefacts?	shortcut scorers; rule-side items; leave-one-kind-out
Does the behaviour carry beyond the generator?	stage 1: TrialGPT, MedEinst, key pairs (negative or not established); stage 2: MedCalc-V, registered criteria (pending)

Table 7: Four validity questions and the evidence for each.

Manifest and provenance. The released manifest gives, for every rule, its source or specification, the text shown in prompts, the program, its applicability semantics, its tests and its version. Requirements that are ambiguous (for example “adequate organ function”) are excluded and not given an invented interpretation. Because every prompt states the rule, correctness is relative to the stated text for all sources. Two limits of the frozen library are recorded with the affected items: for 146 criteria (26 of them in invented rules) the wording of the criterion claim omits the time or person scope that the rule states, and 7 cases contain a general line that contradicts a met condition.

Rendering audit. Before the freeze, 450 rendered groups were read in six model sessions; no person took part. In the first round (300 groups, four sessions) the sessions disputed 11, 9, 10 and 12 labels; after the generator was changed, the second round (150 groups, two sessions) disputed none. Missing-input lines that could be read as “absent” were reworded to “unknown”, and presentation edits that changed a fact, sampled alternatives ruled out by the condition, and near-misses that differed from the flip in more than one attribute were removed. A model reading its own generator’s output can repeat the generator’s error, so this is not a human check of text fidelity. The authors’ reading of 300 groups is **pending**.

Constraint rules and claims. Each rule states a default option, a condition and an alternative. It is derived from a public drug-label or guideline recommendation in three steps taken from Shen et al. (2026), whose own functions are not released: a model extracts the recommendation, writes it as a decision tree, and compiles the tree to a program. We keep a rule if the tree is complete, the program passes the boundary tests and agrees with the tree on sampled inputs, and show the verbalised tree in the prompt. No clinician checks the tree against the source, which is why labels are relative to the stated rule. Every case reports the same fields with raw values, so the presence of a field carries no label information.

Structure and complexity. Rule counts overstate variety, so Table 8 reports structure. Most judgments concern one condition of one rule; where that is so we call the task local applicability verification.

	Train	L2
Rules (fold 1)	223	103
Distinct program structures (names, constants removed)	tbd	tbd
Conditions needed per judgment (mean; max)	tbd	tbd
Dependency depth (mean; max)	tbd	tbd
Cases with competing mentions of the concept (%)	tbd	tbd
Judgments with a numeric comparison (%)	tbd	tbd
Judgments with an exception (%)	tbd	tbd
Criteria under which a past or family finding counts (%)	30.5 (library)	

Table 8: Structural statistics of rule_v1.

Splits and overlap. There are three folds; in fold 1, 223 rules are used for training, 27 for L1 and 103 for L2. The levels change different things: L0 the generated cases only; L1 rule identities and constants; L2 logical compositions; all test sets the wording. Between training and L2 no canonical program and no rule text is shared, and 76.3% of the criterion subexpressions of L2 rules also occur in training rules (72.1% and 85.3% in folds 2 and 3): held-out structure means unseen compositions of seen parts. Rendering templates (1,032 for training, 516 for test) and cue words for relatives, negation and time are disjoint. All cases derived from one base state are in one split. On fold 2 the verdict-only model trained on blocks reaches 63.1% on L2 (fold 1: 58.2%); the other fold-2 and fold-3 runs are **finished and not yet collected**. The set with altered thresholds (L3-alt) does not test what we intended: the untrained backbone solves 83.9% of it and the verdict-only model trained on blocks 94.6%, far above their L2 scores, so it is listed in Table 22 and not used as evidence.

Versions. rule_v1 was frozen, with hashes of every set, before the runs reported here; an independent rebuild reproduced every set byte for byte. Sets built later (rule-side items, corpora for the ablations) are separate named versions, each frozen before any system was scored on it. Of the 400 rule-side items, 38 were found after the freeze to have a known issue (23 on age inclusivity, 15 on the currency of a diabetes diagnosis); results are given with them, and without them in Table 19.

C Triplet Construction

Invariants. Every edit is checked on the target claim by executing the program: a flip must change the claim, a near-miss and a presentation edit must not, also for rules that combine criteria. A re-execution of this check over all frozen sets found no violation in 16,200 groups (203,014 records). Of the proposed groups of the triplet corpus, 1.1% (65 of 5,688) were rejected, all for a presentation

edit; none of the 2,000 proposed L2 groups was. Each training corpus has 60,000 records; L2 has 2,000 triplets and the development set 300. No evaluation input contains the edit type, the role of the case, or the state.

Proposition 1. *Let u be deterministic. (i) If $u(s, x) = g(s)$, then $d(x) = d(x^f)$. (ii) If $u(s, x) = g(x)$, then $d \equiv 0$. (iii) If $u(s, x) = g(s, h(x))$ for a feature map h with $h(x^f) = h(x^n)$, then $d(x^f) = d(x^n)$. In each case the triplet is not solved; under (iii) a correct reversal on the flip implies failure on the near-miss.*

Proof. (i) and (ii) are immediate from the definition of d . (iii) $d(x^f) = g(s, h(x^f)) - g(s', h(x^f)) = g(s, h(x^n)) - g(s', h(x^n)) = d(x^n)$, while a solved triplet needs $d(x^f) < 0 < d(x^n)$. $\square$

Let $h_k(x) \in \{0, 1\}$ indicate whether the decisive concept k is named anywhere in the case, whatever its value, subject, status or time. For a finding that the flip introduces, the base case is rendered without naming k and an applicability near-miss names it, so $h_k(x) = 0$ and $h_k(x^f) = h_k(x^n) = 1$: (iii) applies. For a measured input k is named in all three cases, so $d(x) = d(x^f)$ as in (i). A scorer that uses the case only through h_k thus solves no triplet, although it can reverse on every flip of the first kind (47% of L2 flips). The proposition is a check on the construction and does not exclude other shortcuts, and it does not extend to sampled decisions.

Semantics. *Several mentions.* A criterion uses the mentions its predicate counts; for a measurement **the current value**, for a finding any counted mention with status present. *Numbers.* Thresholds are inclusive or exclusive as the rule text says, and values exactly at a threshold form their own near-miss kind. *Time.* In rule_v1 a mention is current or past, and a past mention carries a year; there is no reference date and no rule with a time window. Windows occur only in the rule-side items and, with a stated reference date and a stated treatment of the boundary day, in the stage-2 sets. *Omission.* Measurements that are not given are unknown; history items that are not mentioned are absent, unless the case says they are unknown. This is a stipulation of the generator, not a general reading of clinical notes. External datasets keep their own conventions; MedCalc-V, for example, scores an unmentioned finding as absent, with documented exceptions.

Missing twins. We distinguish explicit absence, which determines a criterion, from an unknown input. With $C_p(z')$ the non-empty set of admissible completions of a partial state z' , a claim s is *supported* if every completion supports it, *refuted* if every completion refutes it, and *undetermined* otherwise; inconsistent states are excluded. A missing twin is kept only if the target claim is undetermined, which an unknown criterion does not by itself guarantee. A verdict “+” means supported; “−” covers refuted and undetermined claims, which verifiers for retrieval keep apart (Qiu et al., 2026; Kamelhar, 2026). MR is the rate at which both claims of a missing twin are rejected at the threshold that gives 5% false rejection of supported claims on development data. It is 0.0% for Med-PRM, 0.6% for the MedS³ PRM, 14.8% for the FoVer PRM, 7.7% for the untrained critic and 37.8% for the prompted summary, and between 99.2% and 100.0% for every trained model: the marker of a missing input is learnable from the 15% of such records in each corpus, so we report this as a limit of the test and not as evidence of abstention.

Harder set. The harder set is built on the L2 rules and adds long notes with unrelated lines, superseded values (an older value crosses the threshold, the latest does not) and relabelled lines; counts per requirement are **tbd**.

Rule-side items. For a fixed case, two rules differ only in the applicability clause: time window, currency, subject, or boundary inclusivity (100 items each; 9,600 records). Each item has three cases (the contested form, an uncontested positive, an absent finding) and is solved only if all six judgments are right. Scorers that ignore the rule text, never count the contested form or always count it solve no item; the rule program solves all, and uniformly random preferences 1.5%.

Text written without the renderer. A kit of 160 case specifications for cases written by the authors, each to be checked by a second author, is prepared; the cases are **not yet written**, and model-rewritten cases are **not built**. Neither is used in this draft. The tests that the renderer does not determine are therefore the rule-side items and, in stage 2, real notes.

D External Tiers of Stage 1

TrialGPT annotations. The released file has 1,015 patient–criterion annotations over 53 patient

summaries and 103 trials (356 inclusion and 659 exclusion criteria), with expert fields (eligibility label, relevant sentences, a rating of a model’s explanation) and, separately, GPT-4 outputs. The expert labels are: included 150, not included 23, excluded 10, not excluded 484, not enough information 293, not applicable 55. Protocol, fixed before any system was scored: inputs are the patient summary as released and the criterion with its type; no GPT-4 field is ever an input; targets are the expert labels, read as *met*, *not met* or *not enough information* for the criterion as written (**confirm the mapping against the evaluation script**); “not applicable” items are left out; “not enough information” is predicted when neither claim clears an acceptance threshold; adapter, prompts and threshold are chosen on the rule tier and on a patient-disjoint development portion (200 items, 10 patients), and the test portion (759 items, 43 patients) is scored once. Intervals resample patients. The patient summaries are synthetic, so this is an external expert-annotated evaluation, not validation on clinical records.

System	Macro-F1	Acc.	Choice
Always “met”	10.3	18.2	–
Always “not met”	22.6	51.2	–
Always “not enough inf.”	15.6	30.6	–
Criterion-type prior	46.0	66.7	–
Critic	65.7	70.4	92.8
+ prompted summary	68.0	71.4	91.3
+ prompted ledger ^a	54.0	62.3	86.0
Verdict only, blocks	64.8	65.7	92.8
seed 2	68.6	69.2	92.0
Verdict only, triplets	70.8	71.1	92.8
seed 1	70.1	70.6	91.1
Evidence summary, blocks	68.2	69.2	84.4
Evidence summary, triplets	69.2	69.0	87.5
Ledger-RM, blocks	70.9	71.8	85.8
Ledger-RM, triplets	69.1	70.6	87.7

Table 9: TrialGPT test portion (759 items, 43 patients; %, seed 0 unless marked). Choice: accuracy of the preferred claim without the option “not enough information” (**confirm the item set**). ^aLenient parsing. Intervals over patients for macro-F1: critic [61.8, 69.7], prompted summary [64.8, 70.8], verdict only [60.3, 69.2] and [66.3, 75.3], evidence summary [63.9, 71.7] and [65.1, 72.9], Ledger-RM [66.7, 74.6] and [64.3, 73.5]. Malformed reader outputs: 1.9% (Ledger-RM, blocks) and 4.0% (triplets).

On the development portion the difference between Ledger-RM trained on triplets and the critic was +13.7 points of macro-F1 [5.0, 20.6] with 10 patients; the test portion, with 43, gives +3.4 [−2.6, 9.2]. The reader’s quoted sentences match

the sentences the physicians marked with precision 57.8% and recall 61.8% for Ledger-RM trained on blocks (**tbd** for triplets); the expert sentences evaluate evidence selection and do not validate the subject, status or time fields.

MedEinst. We use the released test split (5,383 pairs; the served reference split has 10,609 pairs, the paper states 10,689). Claims are the two diagnoses of a pair, so the task is a two-way choice and not the benchmark’s own evaluation over all diagnoses; our pair accuracy is not comparable with its published robust accuracy. With two options, uniformly random preferences are right on both members of 25% of pairs and a scorer that always prefers the control diagnosis on 0%.

System	Pair	Control	Trap	Same
Med-PRM	16.2	67.5	45.4	80.5
MedS ³ PRM	25.1	79.1	44.2	73.1
FoVer PRM	21.0	71.8	48.1	77.9
GenPRM-7B ^a	22.0	58.0	46.5	–
Critic	24.2	71.8	51.6	75.0
+ default correction	24.7	70.4	53.3	74.3
Verdict only, blocks	20.5	68.3	51.8	79.1
seed 2	18.2	65.2	52.3	81.1
Verdict only, triplets	22.8	69.5	52.9	76.8
seed 1	18.7	67.6	50.8	81.0
Evidence summary, blocks	22.8	61.6	56.1	72.1
Evidence summary, triplets	25.1	73.2	47.3	70.3
Ledger-RM, triplets ^b	15.7	44.4	35.1	–

Table 10: MedEinst test pairs, two-way choice (% seed 0 unless marked). Same: share of pairs whose members receive the same diagnosis, computed from the three preceding columns for systems without ties. ^a200 pairs, 5.5% ties. ^b48.2% ties. Intervals over pairs are about ± 1.2 points (critic [23.0, 25.4], Ledger-RM [14.7, 16.7]); resampling the 134 label pairs widens them to about ± 10 points.

Every system without ties gives the same diagnosis to both members of 70–81% of pairs and is wrong on both in at most 5.1%. The failure of Ledger-RM on this set is one of format: 35.9% of its reader outputs are malformed (3,866 of 10,766), and in an inspection recorded by the evaluating author (**not reproducible from a result file**) 30.2% of reader outputs contained an entry longer than five lines. A model trained on clinical pairs (7,807 MedEinst reference pairs and 122 key pairs from MedQA training questions) and tested on pairs whose diseases were held out (115 pairs) is right on both members of 7.8% and 43.5% of pairs at two seeds, with trap accuracy above 95% and control accuracy of 9.6% and 47.0%: it has learned to

prefer the trap label, and we do not use it.

Key pairs. Pairs are mined within a split and reduced to a one-to-one matching, so that no question occurs in two pairs. Excluded: negatively phrased lead-ins, options that refer to other options, and questions without a case description. Requiring that each question’s keyed answer occurs among the other’s options leaves 13 MedQA and 3 CareQA pairs, too few to score, so the reported sets use **the one-directional rule (state it)**: 357 and 225 pairs. Pair accuracy on MedQA and CareQA: critic 79.8 [75.6, 84.0] and 84.4 [79.6, 88.9]; prompted summary 79.6 and 80.9; verdict only on blocks 78.7 and 82.7, on triplets 78.2 and 83.1; prose pipeline on triplets 33.3 and 55.6; Ledger-RM on blocks 35.9 and 44.0, on triplets 30.5 and 38.2 (ties 25.8% and 38.2%, malformed reader outputs 14.8% and 25.1%); Med-PRM 57.7 and 56.0; MedS³ PRM 39.5 and 49.8; FoVer PRM 67.2 and 69.3. By normalised question text, CareQA shares no question with MedQA.

System	Mean	Control F1	Faithf.	Consist.
Critic	80.9	78.3	84.8	79.5
Verdict only, blocks	85.6	79.2	93.8	83.9
Verdict only, triplets	85.4	81.0	90.6	84.5
Evidence summary, triplets	73.2	65.8	86.0	68.0
Ledger-RM, triplets	tbd	tbd	tbd	tbd

Table 11: NLI4CT-P (% seed 0; 500 control statements). Faithfulness and consistency follow the task definitions.

NLI4CT-P. Both verdict-only models are above the backbone, with and without near-misses, so this gain is not evidence for near-miss supervision; no paired test was planned. The slice whose premise is the eligibility section of a trial, where the premise is a stated rule, is a stage-2 analysis (**tbd**).

Not run. CondMedQA (Parekh et al., 2026) is released without the general answers that our readout needs, EHRNote-ChatQA (Kim et al., 2026) was not available to us, and MedPRMBench (Wu et al., 2026) has no public data release; statements about them in earlier drafts are removed.

E Criterion Grounding: Sources and Protocols

Everything in this appendix is a protocol for stage 2. Dataset facts printed in black were verified on the released files; no stage-2 set has been built and no system scored.

Tier	Criterion and its source	Case text	Label basis
<i>Controlled edits, labels by execution</i>			
Rule triplets	stated rule from our library (353 rules)	generated	rule program
Rule-side items	two rules that differ in one applicability clause	generated	rule program
Class triplets [†]	class-level criterion; membership from RxClass, HPO, Mondo	generated	closure of the class
Knowledge-base triplets [†]	pairwise criterion from the DDXPlus condition–evidence lists	DDXPlus test patients, edited by script	exclusion procedure on the lists
MedCalc-V edits [†]	published score definition shipped with the data	case reports and vignettes; one value or sentence edited	released entity values and rule code
<i>Released labels, criterion stated</i>			
MedCalc-V criteria [†]	published score definition shipped with the data	380 notes, 230 human-written	released entity values and rule code
Registered criteria [†]	trial registry text; structure from Chia and LCT annotation	generated	program compiled from the annotation
TrialGPT	trial criterion text	53 patient summaries	physician criterion-level labels
<i>Released labels, no criterion stated</i>			
MedEinst	none; derived from the DDXPlus lists as an intervention [†]	narrated DDXPlus cases, one sentence edited	benchmark labels
Key pairs	none; self-written as an intervention [†]	exam vignettes	answer keys
NLI4CT-P	trial report section	statements	expert labels, edited statements

Table 12: Tiers, with the source of the criterion and the basis of the label. [†]Not yet built or scored (§6.5). MedCalc-V: the rule-based subset of MedCalc-Bench Verified.

MedCalc-V. MedCalc-Bench Verified, tag v1.0.8, CC BY-SA 4.0. The test file has 1,100 rows for 55 calculators; the 19 rule-based ones (categories risk, severity and diagnosis) have 380 rows, of which 230 have notes extracted from published text (186 PMC case reports, 42 USMLE vignettes, 2 TREC topics) and 150 have notes written by a model and edited or approved by a clinician. The training file has 2,426 rule-based rows (1,669 extracted); the original benchmark paper states that the extractions of its training set were not verified by hand, and we treat them as silver. Eight scores have only extracted test notes (CHA₂DS₂-VASc, CURB-65, Centor, Charlson, FeverPAIN, Glasgow-Blatchford, SIRS, Wells for pulmonary embolism). Each row gives the text of the score, a dictionary of entity values, and the answer. *Rule code:* we implement each score from its stated text; a score enters the tier only if our code reproduces the released answer on every row from the released entities (tbd of 19 scores, tbd rows). *Items.* (A) Criterion claims on untouched notes, in three strata: value stated in the note, finding explicitly denied, entity absent and settled by the benchmark’s default. (B) The numeric values that lie within a narrow band of a threshold without any edit. (C) Single-value edits: for a numeric criterion whose value string

occurs exactly once in the note and feeds no other criterion, the value is replaced by one across the threshold (flip) or by one on the same side and nearer to it (near-miss); labels are recomputed by rule code. (D) One-sentence insertions for a criterion that is not met: the qualifying finding attributed to a relative, negated, or, only for criteria that require a current state, placed in the past (near-misses), and the same finding affirmed for the patient (flip). *What is and is not exact.* Labels are exact given the entity dictionary; the dictionary is released annotation and can be wrong. Of the 62 rule-based test rows whose note also occurs, unchanged, in the independently corrected labels of Ye et al. (2025), that set leaves 10 without an answer and gives a different answer for 9 of the other 52. Edits of types C and D compare cases within one note, so an error in an unrelated entity cancels for criterion claims. Inserted sentences are written by us, as in the rule tier, now inside real notes. History-type criteria (a prior stroke in CHA₂DS₂-VASc) count past findings, so a past mention there is a flip, not a near-miss. *Adaptation.* Verifiers are also fine-tuned on items built from the training notes, with and without near-miss edits at equal budget, with a fixed set of scores held out entirely for test.

Registered criteria. Chia (Kury et al., 2020) annotates 12,409 criteria of 1,000 trials with entities, values, temporal constraints, negation and Boolean scope (inter-annotator κ of 0.80 at phrase level), and its authors state that an annotation graph is meant to be read as a Boolean expression over patient data; the Leaf Clinical Trials corpus (Dobbins et al., 2022) annotates 1,006 trial descriptions with comparison operators, values, units, temporal periods, negation and family members. Both are CC BY 4.0. We keep single-condition criteria that carry a numeric threshold or a time window and none of the corpora’s flags for criteria that cannot be represented or need subjective judgment (about 450 in the Leaf corpus by a first count; to be recomputed), and compile the annotation to a program by script. The annotation fixes structure, not execution semantics: whether “within six months” includes the boundary day, and what the reference date is, remain our stated conventions and are printed in the prompt. Cases are generated as on the rule tier, with dates. Trials and criterion texts that occur in the TrialGPT annotations are excluded. An earlier set of 95 criteria from 85 trials that we formalised ourselves is not scored: its programs were never signed off by two authors.

Knowledge-base criteria. The public part of the DDXPlus knowledge base is two files: for each of 49 conditions the set of associated symptoms and antecedents (888 links, 605 to symptoms and 283 to antecedents), and for each of 223 findings (110 symptoms, 113 antecedents; 208 binary, 10 categorical, 5 multiple-choice) its question text in English. No probabilities are released. For a pair of diagnoses (A , B) a script renders, from the question texts and in a canonical order of the two names, the findings listed for A only, for B only and for both, followed by the procedure: *a finding listed for one diagnosis only excludes the other; if findings of both exclusive lists are present, or of neither, the criterion does not decide*. The script and the hashes of the two files are committed before any model is run. Because, by its authors’ description, the same knowledge base guided the construction of MedEinst’s trap cases, handing the lists to a model changes what that benchmark measures; the claim we test is the contrast between trained and untrained verifiers under identical text. *Controls*: a symbolic executor that maps narratives to findings and applies the procedure, run first; a string-match baseline that counts hits in the two exclusive lists;

every system with and without the criterion; the criterion of an unrelated pair; the pair’s own criterion with the two lists exchanged; the two diagnosis names replaced by neutral labels; lists without the procedure. *Knowledge-base triplets*: for a DDX-Plus test patient with diagnosis A and a second diagnosis B , a script removes the findings listed for A only and adds findings listed for B only (flip), or adds a B -only finding in a form the stated criterion does not count (near-miss); labels follow from the procedure. The lists themselves contain findings about relatives and about the past, which count where listed, so a verifier that has learned “family means ignore” fails here by design.

Class membership from open ontologies. Table 13 lists the sources. A class-level criterion is grounded to a class identifier and a release. *Membership* is the is-a closure in HPO and Mondo; for drugs it is defined at the level of ingredients: a mention is normalised to an RxNorm ingredient, and it belongs to a class if the ATC fourth-level class and the FDA established pharmacologic class served by RxClass both contain it. *Negatives* for the near-miss must pass an assumed-disjointness test in the sense of He et al. (2023): no subsumption in either direction and no common descendant; combination products and ingredients with several class memberships are excluded. *Class triplets*: base, no member mentioned; flip, a member, under its ingredient, brand or synonym name; near-miss, a relative outside the closure, or a term too general to establish membership. Crossing with the other conditions (a relative’s lisinopril, lisinopril stopped last year) is free. *Linking* at inference is retrieval over labels and synonyms of the release followed by a choice among the retrieved candidates by the reader, never free generation of identifiers; its accuracy is reported separately from the verifier’s. Two regimes are kept apart: *open book*, in which the member list or the gate is available, and *closed book*, in which the model must know the class. Labels are exact relative to the release and the membership rule above, both stated with every item. Contraindication assertions (MED-RT, via RxClass) are used only in the positive direction: a missing assertion is not evidence of safety.

What is compared, and what would count against us. On MedCalc-V results are reported for the 230 human-written notes and the 150 model-written ones separately, for criteria whose value the note states and for those settled by the bench-

Domain	Source	Terms of use
Findings	Human Phenotype Ontology	free use with attribution; no alteration
Diseases	Mondo; Disease Ontology as fallback	CC BY 4.0; CC0
Drugs	RxNorm ingredients (prescribable subset, RxNav)	no licence required
Drug classes	ATC level 4 and FDA class through RxClass; MED-RT relations	public API; ATC names are not redistributed
Labels	openFDA and DailyMed label sections	public domain / CC0

Table 13: Open sources for class membership; none needs a SNOMED CT or UMLS licence. **Licence wording to be re-read at release time.**

Tier (metric)	Critic		Ledger-RM-G			
	none	st.	none	st.	self	wrong
<i>Criterion stated by the dataset</i>						
MedCalc-V criteria (acc.)	tbd	tbd	tbd	tbd	tbd	tbd
MedCalc-V edits (TA)	tbd	tbd	tbd	tbd	tbd	tbd
Registered criteria (TA)	–	tbd	–	tbd	–	tbd
TrialGPT (F1)	–	65.7	–	tbd	–	tbd
<i>Criterion derived</i>						
Class triplets (TA)	tbd	tbd	tbd	tbd	–	tbd
KB triplets (TA)	tbd	tbd	tbd	tbd	tbd	tbd
MedEinst (pair)	24.2	tbd	tbd	tbd	tbd	tbd
<i>No source</i>						
Key pairs, MedQA (pair)	79.8	–	tbd	–	tbd	tbd

Table 14: Criterion ladder (%): no criterion in the prompt, the stated or derived one (st.), one written by the frozen backbone from the task identity (self), and a wrong one (roles exchanged or taken from another item). Under a wrong criterion a rule-follower should score low. Black: measured. Class triplets: classes held out from training; “none” is the closed-book condition, “st.” gives the member list or the gate. Accuracy on MedCalc-V criteria is balanced over items that score points and items that score none. A dash marks a condition that is not defined: for registered criteria and TrialGPT the claim is the criterion itself.

mark’s default, and for values that lie naturally near a threshold. The comparison of interest is that of §6.2 on clinical prose: backbone tbd, blocks tbd, triplets tbd (triplet accuracy on edited notes), and the same after adaptation on the training notes with and without edited near-misses at equal budget (tbd, tbd). Registered criteria extend the rule language to time windows with a reference date, the clause on which every current system is weakest: tbd without and tbd with the date check of the gate. On the rule tier, removing the rule text from the prompt gives tbd and exchanging it for the rule of another group tbd. For class-level criteria on held-out classes we

compare the learned verdict (tbd), the learned verdict given the member list (tbd) and the gate (tbd), by depth and frequency of the class, and report linking accuracy separately (tbd). On MedEinst the symbolic executor’s pair accuracy (tbd) and the share of pairs it decides (tbd) bound what any rule-follower can gain. For key pairs the case-visible composite is compared with the verdict-only verifier (tbd against 78.2%). If the ordering of the rule tier does not reappear on MedCalc-V, or the trained verifier is as indifferent to the criterion as the backbone, the claim that near-miss supervision teaches rule-relative selectivity on clinical text fails.

Self-written criteria and controls. For key pairs the frozen backbone writes, from the option set and without the vignette, what would have to hold for each of the two answers; the text is given identically to every system, a mismatched text from another pair is the control, and both rows are reported with the row without a criterion. For every tier the ladder is reported per system with the untrained backbone alongside, so that an effect of the added text on any model is not mistaken for an effect of training. A rule-side invariance arm complements the rule-side items: the rule text is rewritten without changing its program (clause order, placement of the exception, synonyms), which executing both programs certifies, and the excess rate of changed verdicts over a rerun baseline is reported (Weng et al., 2026; Yang et al., 2026) (tbd).

F Ledger Format, Reader Statistics and Gate

```
need: penicillin allergy in the patient
found: "penicillin allergy (anaphylaxis)"
subject: other (mother)
status: present
time: current
```

Given this ledger, the rule and the claim “Order amoxicillin”, the judge returns +. For the main model, malformed ledgers occur in at most 0.2% of rule-tier reader outputs, in 1.9–4.0% on TrialGPT, 14.8–25.1% on key pairs and 35.9% on MedEinst: the format degrades as the input moves away from a stated rule and a short case. A malformed ledger is recorded as an invalid outcome and both claims are rejected. Neither record format may state a verdict: no claim sentence and no decision phrase occurs in the reader outputs of any of the 27 two-stage runs (for example, in none of 37,497 summaries of one run). In 30.2% of judge training examples the ledger shown is a donor’s. A verification pass that

re-reads every entry against the case and regenerates rejected ones once adds 0.6 points ($[0.1, 1.2]$, $p = 0.012$). In triplet and pair evaluations one ledger serves both claims.

Record for stage 2. The stage-2 record adds what the stage-1 results showed to be missing: the date expression, so that a window can be computed, an optional normalised concept, and the applies bit.

need: stroke within 6 months before 2025-03-01
found: "ischaemic stroke in June 2024"
subject: patient
status: present
time: past (2024-06)
applies: no

Condition	Check	Not computable when
quotation value	string occurs in the case comparison after unit conversion	never no threshold in the criterion; no number in found
time	date arithmetic against the reference date	no window; no date expression; no reference date
concept	mention linked to an ontology term; term in the closure of the class	criterion names no class; linker abstains
subject, status	set membership on typed fields	scope not stated in the criterion

Table 15: Checks of the symbolic gate. If every applicable check is computable, their conjunction replaces the reader’s bit; otherwise the bit stands and the judge also sees the case.

G Diagnostics

Parts and whole. For each base–flip pair we build a *reading* pair, whose claims quote the decisive finding, and an *application* pair, which scores the original claims against a one-line case with only that finding. With P , K , U the events of a correct reversal on the reading pair, the application pair and the full pair, the composition gap is $G = 1 - \Pr[U \mid P \wedge K]$; it is descriptive, since the tasks differ in length and in what must be selected. For the untrained backbone on 1,000 pairs, $P = 97.7\%$, $K = 96.0\%$, $U = 88.9\%$, and $G = 7.9\%$ among the 944 pairs with P and K . On L0 its base accuracy is 68.6% and, among triplets with a correct base preference, it reverses on the near-miss in 34.0% ($n = 686$). Other signals are **tbd**.

Shift relative to a missing-input baseline. Let $d_0 = d(x^m)$, $\delta(x) = d(x) - d_0$, and $y \in \{+1, -1\}$

Scorer (L2)	Rev	Hold	TA
Always default	0	100	0
Claim only	0	—	0
Concept named (h_k)	47	0	0
Bag of words, case	tbd	tbd	7.0
Attribute-blind, logistic	tbd	tbd	40.0
Trigger lexicon, training cues	tbd	tbd	54.0
Trigger lexicon, test cues	tbd	tbd	99.5
Clinical trigger tagger	96.7	81.0	77.7

Table 16: Shortcut scorers on L2 (%). Attribute-blind: a logistic model on the decisive concept’s mentions without subject, status and time; it solves numeric triplets only. Trigger lexicon: a rule-based reader with negation, experiencer and temporality triggers in the style of ConText (Harkema et al., 2009), followed by the rule program; given the test cue phrases it is a reference and not a baseline. Clinical trigger tagger: the default rules of a frozen clinical toolkit, followed by the rule program. Trained models that exceed these scorers outperform the shortcuts tested here; this does not show that they use no shortcut.

System	thr.	neg.	num.	subj.	time
Verdict only, blocks	33.8	93.2	92.0	11.5	60.5
Ledger-RM, blocks	72.0	97.5	99.8	48.5	58.8
Verdict only, triplets	93.2	91.2	92.8	91.0	88.2
Ledger-RM, triplets	97.8 or above for every kind				

Table 17: Triplet accuracy on L2 by near-miss kind (%; seed 0): value at the threshold, negated finding, value near the threshold, other person, past finding. Denominators per kind, and near-miss correctness given a correct base judgment, are **tbd**.

	Verdict		Prose		Ledger-RM	
Removed kind	out	in	out	in	out	in
Other person	7.0	91.0	89.2	93.2	62.5	97.8
Past	60.2	88.2	61.5	99.8	86.2	100.0
Threshold	49.8	93.2	99.0	100.0	87.0	100.0

Table 18: Leave one near-miss kind out (%; seed 0). Out: triplet accuracy on L2 triplets of the kind that was removed from training; in: the same triplets for the model trained on all kinds. Paired differences range from -84.0 $[-89.2, -78.2]$ (other person, verdict) to -1.0 $[-3.2, 0.0]$ (threshold, prose). Triplet accuracy on all L2 triplets with the kind removed (other person, past, threshold): verdict 76.2, 86.1, 82.8; prose 97.2, 91.5, 99.3; Ledger-RM 91.2, 96.8, 96.5.

the correct sign of $d(x)$. For cases with $y d_0 < 0$, where the baseline preference opposes the label, the preference is right if and only if $y \delta(x) > |d_0|$. We sort these cases into four disjoint classes, in this order: *crossed* (the inequality holds); *unmoved*

System	XA	XA ⁻	window	curr.	subj.	incl.
Med-PRM	13.5	14.6	12	16	10	16
MedS ³ PRM	14.0	15.2	16	17	17	6
FoVer PRM	44.5	46.7	42	66	51	19
Critic	39.0	42.5	35	42	66	13
+ default correction	58.5	59.9	38	74	93	29
+ prompted summary	80.5	83.1	65	88	98	71
Generated-program verifier	82.8	83.1	67	93	88	83
Verdict only, blocks	60.8	tbd	44	75	47	77
Verdict only, triplets	90.2	tbd	66	100	95	100
Evidence summary, triplets	88.5	tbd	54	100	100	100
judge also sees the case	91.5	tbd	72	96	98	100
Ledger-RM, triplets	89.8	tbd	68	93	98	100
+ decision field	98.8	tbd	95	100	100	100
reader writes bit only	97.8	tbd	91	100	100	100
condition derived by the reader	86.5	tbd	49	99	98	100
— ledger resampling	78.8	tbd	55	96	64	100

Table 19: Rule-side items (% , seed 0): crossed accuracy on all 400 items, without the 38 items with a known issue (XA⁻), and by the clause that differs between the two rules (100 items each). Over five seeds the window column ranges from 43 to 71 for the verdict-only model and from 67 to 76 for Ledger-RM, both trained on triplets.

($|\delta| \leq \tau$, the model’s median $|d(x^p) - d(x)|$ over presentation edits); *wrong direction* ($y\delta < 0$); *short*. The shift ratio κ divides the median of $y\delta(x)$ over these cases by the model’s median response to an error inside a step (state the pairs used for the denominator; the planned source, MedPRMBench, is not released). Untrained backbone: 43.1% crossed, 17.3% short, 23.6% unmoved, 15.9% wrong direction, $\kappa = 0.50$. Med-PRM: 21.8% crossed, 40.9% short, 31.2% unmoved, $\kappa = 0.71$. Verdict only trained on blocks: 95.8% crossed, $\kappa = 1.01$. Med-PRM’s preference moves in the right direction and stops short in two of five conflicting cases.

Readouts and sampling noise. Dedicated PRMs use their released scoring head and local models the log-odds of the verdict token. For the untrained backbone the triplet accuracy on L2 is 27.1% in the full-precision scoring run and 25.4% from the log-odds of a second pass that returns the top tokens; sampling the verdict ten times gives 24.8% (s.d. 0.4) at temperature 0.7 and 24.0% (s.d. 0.7) at temperature 1.0. Readout and precision thus move this number by up to three points, which is small against the differences in Table 2.

Default correction. The weight α was chosen on development triplets, where triplet accuracy is 39.7% at $\alpha = 0$, 47.3% at 1, 50.7% at 4, 52.0% at 16 and 53.0% in the limit, which uses only the difference from the score without the case; the limit was selected after the grid had been extended on development data. On L2 this gives 26.4%

[20.3, 32.7], against 27.1% without correction.

Exploratory. A linear probe on the residual stream of the untrained backbone, trained on L0 and tested on L2, predicts the criterion outcome with 86.9% accuracy (control task (Hewitt and Liang, 2019): 69.8%), and with 94.5% against 41.3% on conditions not seen by the probe; decodability does not show use (Elazar et al., 2021).

H Additional Results

Systems. *Released reward models:* Med-PRM (Yun et al., 2025), the MedS³ PRM (Jiang et al., 2025), the FoVer PRM (Kamoi et al., 2026), ThinkPRM-14B (Khalifa et al., 2025) and GenPRM-7B (Zhao et al., 2025); the last two are scored on fixed subsets so far. MedPRMBench has no public data or model release, so we could not train a step-error PRM on it, and the step check throughout is the released Med-PRM. *Training-free* on Qwen3.5-9B: the zero-shot critic; default correction, $u(s, x) + \alpha [u(s, x) - u(s, \emptyset)]$ (Shi et al., 2024); a prompted two-stage summary and a prompted ledger; and a verifier that writes and executes a program for the stated rule (Ocampo Osorio et al., 2026). *Trained comparisons* on Qwen3.5-9B: the five representations on the four corpora; verdict only on FoVer’s formal data; a pairwise loss; a linear concept scorer; and probe re-weighting, our adaptation of Liu et al. (2026a) to case edits. *References* that use a rule program at test time: a general-purpose trigger tagger (Harkema et al., 2009) followed by the program, and the program applied to the trained reader’s ledger. Larger open and closed judges (Qwen3.5-27B, Llama-3.3-70B, Kimi K3, GPT-5.4, Gemini 3.1 Pro, Claude Opus 5.5) are not yet scored. All four training corpora contain 15% missing-input and 15% presentation records; the loss is computed per example, so the corpora differ in their cases and not in the objective or the batching. In the natural corpus the rate of the decisive condition is a parameter of the generator, not an empirical prevalence. Neither record format of the two-stage systems contains a verdict (Appendix F).

Further ablations. *Finding the condition.* When the reader must derive the condition under test from the rule and the claim, triplet accuracy falls to 86.5% ($-12.7 [-16.8, -8.9]$): near-miss correctness stays at 98.6% and reversal falls to 87.0%. *Rule-side items.* Blocks-trained models solve 35.8–

System	Rule tier				Criterion stated		No criterion stated					
	L2	L3-inv	XA	Hold	TrialGPT		MedEinst			Key pairs		NLI4CT-P
					F1	choice	pair	control	trap	MedQA	CareQA	avg.
<i>Training-free, Qwen3.5-9B</i>												
Critic	27.1	33.4	39.0	32.2	65.7	92.8	24.2	71.8	51.6	79.8	84.4	80.9
+ default correction	26.4	tbd	58.5	tbd	tbd	tbd	24.7	70.4	53.3	tbd	tbd	tbd
+ prompted summary	67.5	tbd	80.5	89.8	68.0	91.3	tbd	tbd	tbd	79.6	80.9	tbd
+ prompted ledger (lenient parse)	59.5	tbd	52.5	87.8	54.0	86.0	tbd	tbd	tbd	tbd	tbd	tbd
Generated-program verifier	89.8	tbd	82.8	94.7	—	—	—	—	—	—	—	—
<i>Trained on rule data only (all clinical columns are zero-shot)</i>												
Verdict only, FoVer data	38.5	tbd	67.0	44.1	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd
Verdict only, rule blocks	58.2	62.9	60.8	61.5	64.8	92.8	20.5	68.3	51.8	78.7	82.7	85.6
Verdict only, rule triplets	91.3	96.9	90.2	97.7	70.8	92.8	22.8	69.5	52.9	78.2	83.1	85.4
Rationale, rule triplets	91.0	96.5	92.8	99.6	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd
Evidence summary, rule blocks	67.3	73.6	80.2	70.1	68.2	84.4	22.8	61.6	56.1	tbd	tbd	tbd
Evidence summary, rule triplets	98.0	97.6 ^a	88.5	99.3	69.2	87.5	25.1	73.2	47.3	33.3	55.6	73.2
Ledger-RM, rule natural	77.2	tbd	86.0	77.8	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd
Ledger-RM, rule balanced	49.4	tbd	54.5	51.6	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd
Ledger-RM, rule blocks	75.3	70.1	76.0	76.2	70.9	85.8	tbd	tbd	tbd	35.9	44.0	tbd
Ledger-RM, rule triplets	99.2	97.9	89.8	99.95	69.1	87.7	15.7	44.4	35.1	30.5	38.2	tbd
<i>Trained with clinical pairs (MedEinst reference pairs and MedQA-train key pairs)</i>												
Ledger-RM, clinical pairs only	77.5	tbd	35.5	85.0	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd
Ledger-RM, rule triplets + clinical pairs	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd	tbd
<i>References that use the rule program at test time</i>												
Trigger tagger + program	77.7	—	—	81.0	—	—	—	—	—	—	—	—
Program on the trained reader’s ledger	99.4	—	—	100.0	—	—	—	—	—	—	—	—

Table 20: All rows of Table 4 (% , seed 0; ^aseed 1). Columns as there. The run that adds clinical pairs to rule triplets was queued and is not finished.

82.2% over formats and seeds. Windows are the weakest clause for every trained system that does not write a decision bit (43–76% over formats and seeds). Training without donor ledgers lowers crossed accuracy from 89.8% to 78.8% while the L2 score stays at 98.8%. *One stage against two.* The rationale’s deficit against the two-stage ledger lies in reversal (91.3%), not in holding (99.6%); the case-visible prose judge is 0.8 points above the case-blind one ([0.1, 1.5]) and statistically level with the ledger (−0.4 [−1.1, 0.3]); on blocks the ledger is 2.7 points above prose over four seeds ([−0.1, 5.6]); the ledger is 13.5 points above the value ledger on triplets ([11.3, 15.9]) and 4.6 on blocks ([1.3, 7.6]). With program-supplied ledgers the judge solves 99.8% of L2 triplets: 12 errors of the full model are rescued, none is introduced and 4 remain. *Corpora without near-misses.* Among the natural, balanced and block corpora the differences between cells at seed 0 (43.0–77.2%) are of the size of the seed-to-seed variation of the block cells (s.d. 5.4–10.7), so we do not interpret them. *Scale.* With the 4B model of the same family the verdict-only model trained on blocks reaches 49.9%, against 58.2% for the 9B model at the same seed; the other cells are not run, so we draw no conclusion about scale. *Formal data.* Verdict only trained on FoVer’s formal data solves 38.5% of L2 triplets, with base accuracy at chance

(49.3%). *Other signals.* ThinkPRM-14B, a general verifier that reasons before it scores, solves 87.5% of 200 L2 triplets; a logistic model on attribute-blind concept–value features solves 40.0%, and a frozen general-purpose clinical trigger tagger followed by the rule program 77.7%.

Objective. Let $N(x)$ be the other training cases of the same rule and condition as x , including its twins, and let $q(\cdot \mid x, \rho, \omega)$ put mass $1 - p$ on $e^*(x, \rho, \omega)$ and spread p uniformly over $\{e^*(x', \rho, \omega) : x' \in N(x)\}$. Then

$$\mathcal{L} = \mathbb{E}_{(x, \rho, s)} \left[-\frac{1}{|e^*|} \log p_\phi(e^* \mid x, \rho, \omega) - \lambda \mathbb{E}_{e \sim q} \log p_\phi(y_\rho(e, s) \mid e, \rho, s) \right], \quad (3)$$

with $|e^*|$ the number of target tokens. Donor ledgers are gold ledgers, whereas at test time the judge reads generated ones; the program-supplied ledger analysis measures that gap.

Training. LoRA rank 64, batch size 64, one epoch over 60,000 records, learning rate **1e-4**, $\lambda = 1.0$, donor probability $p = 0.3$ (a donor’s ledger in 30.2% of judge examples). The two loss terms are separate supervised examples in one corpus: one reader example per case and condition, shared by both claims, and one judge example per claim. A deterministic parser maps ledger text to a

Variant (rule triplets only, seed 0)	Rev	Hold	TA	XA
Ledger-RM (facts only)	99.2	99.95	99.2	89.8
+ decision field	99.7	99.8	99.5	98.8
decision field only	99.5	99.9	99.3	97.8
program on predicted bit	99.7	99.8	99.5	–
reader writes bit only	99.95	100.0	99.95	97.8
condition derived by the reader	87.0	98.6	86.5	86.5
+ verification pass	99.8	100.0	99.8	91.5
program-supplied ledger	99.8	99.95	99.8	–
program on predicted ledger	99.4	100.0	99.4	–
One-stage ledger (rationale)	91.3	99.6	91.0	92.8
Value ledger	99.1	86.6	85.7	70.5
Evidence summary (matched prose)	98.5	99.3	98.0	88.5
judge also sees the case	99.0	99.8	98.8	91.5
Concept scorer	34.4	58.4	26.8	12.0
Premise gate	tbd	tbd	tbd	tbd
– near-misses	98.7	76.2	75.3	76.0
+ probe re-weighting	98.6	56.0	55.3	25.8
– other-person near-misses	97.5	93.5	91.2	90.8
– past near-misses	99.5	97.2	96.8	84.2
– threshold near-misses	99.2	97.3	96.5	85.8
– presentation edits	99.3	99.3	98.7	90.0
– ledger resampling	98.8	100.0	98.8	78.8
Verdict only, pairwise loss	91.8	98.2	91.4	92.5

Table 21: Ablations on L2 and on rule-side items (%). Decision field: the reader also writes whether the condition applies; decision field only: the judge sees that bit and the claim. Condition derived: the reader is given the rule and the claim and must find the condition under test. Verification pass: every entry is re-read against the case and regenerated once if rejected, after Agand (2026). Program-supplied ledger: the judge receives the target ledger. Program on predicted ledger or bit: the rule program replaces the judge and is used at test time. Concept scorer: fields predicted by linear heads and a linear verdict. Premise gate, after Wang et al. (2026): not run. Probe re-weighting: near-misses are not trained on; base-flip pairs whose preference shifts under near-miss and presentation edits are down-weighted, our adaptation of Liu et al. (2026a). Rows that remove one near-miss kind show all L2 triplets; Table 18 shows the removed kind. Paired differences to the first row: verification pass +0.6 [0.1, 1.2]; bit-only reader +0.8 [0.3, 1.4]; condition derived −12.7 [−16.8, −8.9]; pairwise loss −7.8 [−12.3, −4.0]; concept scorer −72.5 [−76.0, −68.4].

System	Dev	L0	L1	L2	L3-inv	Hard	Re-apply	L3-alt
Critic	39.7	44.4	42.8	27.1	33.4	24.6	61.1	83.9
Verdict only, blocks	67.3	68.8	73.8	58.2	62.9	59.1	77.7	94.6
Ledger-RM, blocks	70.3	73.5	78.2	75.3	70.1	76.4	76.0	85.3
Verdict only, triplets	99.7	99.0	97.6	91.3	96.9	90.5	99.4	100.0
Rationale, triplets	98.0	98.1	97.7	91.0	96.5	90.4	99.3	100.0
Evidence summary, triplets ^a	98.7	98.4	97.4	98.1	97.6	98.1	98.8	100.0
Ledger-RM, triplets	99.7	99.7	99.1	99.2	97.9	98.3	99.7	100.0

Table 22: Rule ladder (triplet accuracy, %, seed 0; ^aseed 1). Dev: trained rules, test wording; L3-inv: invented rules; Hard: L2 rules with long notes and superseded values; Re-apply: reading and application pairs; L3-alt: altered thresholds, which the untrained backbone already solves in 83.9% of cases (Appendix B).

Cell	Seed					Mean (s.d.)
	0	1	2	3	4	
<i>L2 triplet accuracy</i>						
Verdict only, blocks	58.2	53.5	60.0	47.6	49.0	53.7 (5.4)
Verdict only, triplets	91.3	91.8	92.2	92.2	91.8	91.8 (0.4)
Evidence summary, blocks	67.3	57.9	72.2	62.2	tbd	64.9 (6.2)
Evidence summary, triplets	98.0	98.1	98.0	98.9	99.5	98.5 (0.7)
Ledger-RM, blocks	75.3	59.5	74.9	60.6	84.5	71.0 (10.7)
Ledger-RM, triplets	99.2	99.0	99.2	99.2	99.7	99.2 (0.3)
<i>Crossed accuracy on rule-side items</i>						
Verdict only, blocks	60.8	47.2	66.2	40.2	37.8	
Verdict only, triplets	90.2	91.2	90.0	87.2	85.5	
Evidence summary, blocks	80.2	51.0	62.5	35.8	tbd	
Evidence summary, triplets	88.5	92.5	90.2	89.2	91.8	
Ledger-RM, blocks	76.0	37.8	71.0	44.5	82.2	
Ledger-RM, triplets	89.8	88.8	91.5	94.0	85.0	

Table 23: Seeds (%). The pooled 95% interval of Ledger-RM trained on triplets is [98.9, 99.6]. Without near-misses the outcome of a run depends strongly on its seed; with them it does not.

Comparison	Isolates
Verdict vs. one-stage ledger	intermediate supervision
One-stage vs. two-stage ledger	separating reading from judgment
Two-stage ledger vs. matched prose	typed structure
Applicability ledger vs. value ledger	subject, status and time
Case-blind vs. case-visible judge	restricting access to the case
Full record vs. one bit	what the judge needs
Own vs. program-supplied ledger	what better extraction could recover
Learned judge vs. rule program on the same ledger	what the neural judge adds

Table 24: Comparisons among interfaces and what each isolates.

state for y_ρ ; incomplete or inconsistent ledgers have no state and yield no judge example. A verdict-only run takes about 2.2 GPU-hours and a two-stage run about 2.8, including evaluation. Unique cases, reader and judge targets, tokens and optimiser steps per configuration are **tbd**; equal record counts do not by themselves mean equal supervision. Scores from the served model used for selection agree in sign with those of the training framework on 99.96% of 8,000 preferences.

Policy training. GRPO from a Qwen3.5-4B policy on rule tasks from training rules, 1,000 steps, one seed, evaluated every 200 steps and at the end on 150 held-out L2 pairs with the rule program. Outcome reward: pair accuracy 88.7% to 96.7%, triplet accuracy 86.0% to 93.3%. Reference-graph reward: 88.7% to 98.0% and 86.0% to 96.7%. Med-PRM as reward: 88.0% to 38.7% and 85.3% to

24.0%, with accuracy over all cases falling from 93.3% to 59.8%; at step 1,000 the mean reward is 0.9997 and pair accuracy measured during training 37.3%. Under the outcome reward only 4.5% of rollout groups have any reward variance, so most groups give no gradient; under Med-PRM 85.3% do, and within groups the reward’s correlation with the outcome is -0.05 on average. What the collapsed policy writes, and whether its errors are failures to reverse, is **tbd**. The two ledger rewards are **tbd**.

Selection	Accuracy		Key		MedEinst	
	MQA	CQA	pair	pair	ctrl	trap
Single sample	82.9	87.0	70.3	22.0	71.6	44.8
Self-consistency	87.8	89.4	78.7	20.6	73.2	45.8
Step check	86.2	89.0	77.9	22.6	74.6	46.0
vignette swapped ^a	85.6	88.1	76.8	23.0	71.8	49.0
Ledger score	83.8	88.4	71.1	26.2	73.2	49.8
Combined (minimum)	85.3	88.9	76.8	22.6	73.8	46.4
vignette swapped ^a	85.6	88.1	76.8	23.0	71.8	49.0
no near-misses	85.1	88.9	76.5	22.0	74.4	45.6
product	86.1	89.2	78.2	22.6	74.6	46.0
fitted logistic	86.1	88.9	77.9	22.6	74.6	46.0
Oracle selection	97.5	97.1	94.7	59.4	87.4	72.0

Table 25: Reranking 16 samples of a frozen policy (**Qwen3.5-9B**; **confirm**) on identical pools (%): 1,273 MedQA and 1,000 CareQA questions, 357 key pairs, 500 MedEinst pairs. Step check: Med-PRM. Ledger score: Ledger-RM trained on rule triplets. Bold: best selector per column. ^aThe two swapped rows are identical in every column, which a correct implementation does not guarantee; **under check**.

Selection protocol. Table 25 gives every selector. Combined minus step check is -0.9 points on MedQA $[-2.1, 0.2]$, -0.1 on CareQA $[-1.5, 1.4]$, -1.1 on key pairs $[-3.4, 1.4]$ and 0.0 on MedEinst pairs $[-1.4, 1.4]$. Training the ledger with or without near-misses makes no difference here ($+0.2$ $[-0.9, 1.3]$ on MedQA). The ledger score alone gives the highest pair accuracy on MedEinst (26.2% against 22.0% for a single sample); no paired test of that difference was planned, and we do not interpret it. The minimum of two calibrated scores is a heuristic: it is an upper bound on the joint probability, not that probability, and scaling each score separately does not account for dependence between the two checks. Pools hold 16 samples per question from the frozen policy; a trace is eligible if it contains a final answer (96.6% of MedQA samples). Platt scaling and the logistic combination are fitted on a MedQA development pool of

359 questions from which key-pair questions are removed. There the AUC of the trace score for answer correctness is 0.654 for Med-PRM, 0.510 for Ledger-RM trained on triplets and 0.479 trained on blocks, and the calibration slope of the ledger score is 0.006 against 0.66 for Med-PRM. On 150 key pairs with nested pools of 1, 16 and 64 samples, pair accuracy is 67.3, 78.0 and 76.0 for the combined score; 67.3, 79.3 and 78.0 for the step check; 67.3, 71.3 and 69.3 for the ledger score; 67.3, 79.3 and 80.0 for self-consistency; and 67.3, 94.0 and 96.7 for the oracle. The curve over the number of training rules of new and of seen structures is **tbd**.

I Extended Related Work

Validity of evaluators. Chen et al. (2026a) define a judge’s validity as invariance to construct-preserving edits together with sensitivity to construct-changing ones, and Weng et al. (2026) hold the judged trajectory fixed and rewrite the policy given to the judge. Yang et al. (2026) show that the effect of a targeted counterfactual edit must be compared with that of meaning-preserving edits of the same size; our near-miss and presentation edits are such baselines with known labels. Robustness work on reward models targets response-side artifacts (Liu et al., 2025a; Zang et al., 2025; Xu et al., 2025). In medicine, answer-changing patient facts (Hou et al., 2026; Chen et al., 2026b; Kim et al., 2025) and clinically irrelevant distractor statements (Vishwanath et al., 2025) have each been tested on answering models; none of these puts a decisive edit and an inapplicable mention of the same concept in one item, varies the applicability clause of a rule with the case fixed, or applies patient-side edits to released medical PRMs.

Supervision with and without label-preserving edits. Minimal label-flipping edits serve training (Kaushik et al., 2020) and evaluation (Gardner et al., 2020; Ribeiro et al., 2020; Thrush et al., 2022), with the caveat that they teach which feature matters and not when it does not (Joshi and He, 2022). Adding label-preserving edits of the same feature removes over-triggering for negation scope (She et al., 2023) and for safety policies: a guard model trained on label-changing policy edits alone reaches precision .50 at recall 1.00, and .90 and .78 with both kinds (Liu et al., 2026b). Natural text remains hard after such training (Ravichander et al., 2022). CROME trains reward models with augmentations that change causal attributes of a

response and with neutral ones (Srivastava et al., 2026); DynaCF down-weights pairs whose margin shifts under content-preserving rewrites (Liu et al., 2026a), which we adapt to case edits as an alternative to training on near-misses; CSR regularises a solver to change its answer when an operator in its own trace is swapped (Shihab et al., 2025). MedGuideX post-trains a clinical policy on factual and counterfactual questions from executable guideline functions and keeps only interventions that change the outcome (Shen et al., 2026). Sensitivity together with invariance is therefore established as a principle. Our negatives differ in kind: they keep the decisive concept in the text and remove one condition of its applicability, a case that Winston (1970) called a near miss, and we measure what a verifier learns without them.

Process rewards and symbolic verification.

PRMs are trained on human step labels (Lightman et al., 2024), rollout estimates (Wang et al., 2024) or generated verification reasoning (Khalifa et al., 2025), which GenPRM extends with executed code (Zhao et al., 2025). FoVer trains step verifiers on labels from formal tools and shows transfer to informal tasks (Kamoi et al., 2026); generating supervision by formal verification is established and is not our claim. NS-PRM separates what a deterministic verifier can decide (schema, arithmetic, units) from what a PRM must learn, trains the PRM on negatives that pass the verifier, and searches verifier-first (Zi et al., 2026); its perturbations all change the label. We adopt the division of labour and add the label-preserving side. Other work injects errors into verified chains (Chi and Wang, 2026) or uses a deterministic verifier as the reward (Pronesti et al., 2026). In medicine, Med-PRM and Med-TIV condition a verifier on retrieved passages (Yun et al., 2025; Zhang et al., 2026), MedCEG rewards coverage of a reference evidence graph (Mu et al., 2025), and MedPRMBench injects 14 error types into chains (Wu et al., 2026). Scoring from a record written in a separate pass follows Chain-of-Verification (Dhuliawala et al., 2024) and EVPV, which extracts premises once per problem and attenuates the reward of steps they do not support (Wang et al., 2026); mediating a reward through explicit concepts precedes our record (Koh et al., 2020; Laguna et al., 2025). Its fields are the attributes that clinical NLP assigns to mentions: negation, experiencer and temporality (Chapman et al., 2001; Harkema et al., 2009; Uzuner et al.,

2011). For composite neuro-symbolic systems in medicine, Wong et al. (2026) name the failure we measure at the reader: an extraction error never activates the rule layer.

Stated rules and where criteria come from.

Reasoning from a stated rule and a described case has a long line of benchmarks (Saeidi et al., 2018; Holzenberger et al., 2020; Clark et al., 2020; Sun et al., 2024), and policy-conditioned guard models are trained to follow rules given at inference (Hoover et al., 2025; Liu et al., 2026b). Whether a verdict depends on the rule has to be tested: deleting, shuffling or replacing the rule left the accuracy of every compliance detector audited by Sadhu et al. (2026) unchanged. For clinical criteria, executable guideline functions are called at inference (Cao et al., 2026), diagnostic hypotheses are checked against guideline-derived disease profiles (Cao and Chen, 2026), conditional QA gates answers on a stated patient condition (Parekh et al., 2026), and trial eligibility is decided criterion by criterion by prompted models (Jin et al., 2024) or by an SMT solver over compiled criteria and typed patient facts, with an audit of whether changing pivotal facts changes the decision (Zhou et al., 2026). As far as we could determine, these systems prompt their reader, and none trains it on inapplicable evidence. Ontologies have supplied exactly labelled subsumption data with sibling negatives (He et al., 2023, 2024) and sibling substitutions that a clinical verifier must not accept (Qu and Färber, 2026); injecting an ontology into weights changed medical QA little (Liu et al., 2025b), whereas a fixed type vocabulary used as a gate before a learned judge helped knowledge-graph QA (Shan et al., 2026). We use ontologies in the second way, and pair a class member that must flip a verdict with a non-member that must not.

Context against prior, and reward exploitation.

How language models arbitrate between memory and conflicting context is well studied (Xu et al., 2024; Wu et al., 2024; Shi et al., 2024), and clinical-trial NLI has been probed with label-preserving and label-inverting interventions (Jullien et al., 2024); claim-only scorers are partial-input baselines (Poliak et al., 2018). Policies trained against learned rewards exploit them (Skalse et al., 2022; Gao et al., 2023; Helff et al., 2026): against math PRMs the reward approaches its maximum while accuracy stays below 4% (Tiwari et al., 2026), rubric and claim-checker rewards in medicine are exploited as

training proceeds (Mahmoud et al., 2026; Ji et al., 2026), and Med-PRM as a process reward gave lower accuracy than an outcome reward in risk-of-bias assessment (Pronesti et al., 2026). Our policy experiment adds a held-out, exactly labelled evaluation and connects the exploited reward to its triplet profile.

J Development History and Error Analysis

[Authors: this section has to be written from your own records. The project log supports the statements below and no others; an earlier draft cited a pilot result for which no record exists, and it is removed.]

What the frozen generator changed. `rule_v1` holds out signature classes, templates and cue phrases; contains rules under which past or family findings count (30.5% of criteria), so that “past or family means ignore” is not a sufficient strategy; adds presentation edits and missing inputs to every corpus; and was frozen before the runs reported here. Before the freeze, a dry run of the attribute-blind logistic scorer solved 69% of triplets, because near-misses then differed from flips in more than one attribute; after near-misses were made to mirror the flip it solves 40.0% on the frozen test (decision log of 2 October; the dry-run number is not a result on the frozen data). Rule-side items test dependence on the rule directly and were built after the freeze as a separate set.

What the results changed. Three expectations of the first plan were wrong and are reported as such: that balanced outcomes would teach reversal (§6.2), that the ledger and the step check would combine into a better selector (§6.6), and that training on rules would transfer without a stated criterion (§6.4). The typed ledger, which the first plan treated as the method, turned out to matter less than the supervision and to be replaceable by one bit. Stage 2 is the plan that follows from these outcomes.

Errors of the trained systems. Sheets with 200 failures of five trained systems on L2 (175 distinct triplets), to be coded by the authors as extraction error, applicability error, unseen wording, rule misread or other, are prepared; the coding is pending.